Machine Translated by Google

AI Basic Society: A Major Transformation in Education

Designing the Future of AI Education 2.0

한국인공지능교육학회 인공지능교육연구소

Machine Translated by Google

Who will survive the competition between education and technology?

"People Growing with AI, a Society Living with AI"

"We are now at the dawn of a new civilization where we will live with artificial intelligence." The leap forward of generative AI

Artificial intelligence is not just about technological advancement; it's raising fundamental questions about how humans think, learn, and live.

is now emerging as an 'intelligent partner' that goes beyond a simple tool and creates, judges, and even influences decision-making.

What role should education play in this changing world?

Traditional AI education has focused on how to use AI. Now, however, we need to think with AI, solve problems, and learn better.

We must shift to education that builds society. We must move beyond a functional understanding of technology to encompass collaboration between humans and technology, and a technological understanding of technology.

We are at a point where we must pay attention to the social values that must be realized.

With this understanding, this report proposes the "AI Education 2.0 Model." This model goes beyond simple technical learning and encompasses "AI understanding education, AI

Through three types of education, including "collaborative education, AI value education," we are instilling the thinking and practical skills necessary for a society where AI is the foundation within education.

We aim to foster AI in an integrated manner. In particular, we aim to foster AI in a way that is based on the AI thinking structure of 'abstraction – automation – autonomy' with 'thinking like AI' as the central concept.

We are exploring an educational approach.

This report presents new directions and concrete practical measures to teachers, educational administrators, policy makers, and researchers in the field of education.

Let's consider a practical model for fostering a human-centered AI-based society and cultivating good citizens who can coexist with AI.

We hope that this report, which proposes and provides suggestions, will serve as a starting point for a better future education.

All researchers

Machine Translated by Google

index

01. Introduction to AI Education 2.0 05

1. Why 'AI Education 2.0'? 06
2. Goals and Targets of 'AI Education 2.0' 08

02. Core Principles of the AI Education 2.0 Model 10

1. The foundation of the AI Education 2.0 model: It must start with digital. 11
2. The singularity of education considering AGI and ASI: Education designed with the future of technology in mind. 12
3. Why we need to understand AI 'properly' above all else: Proper cooperation and value come from understanding 13
4. New Thinking Skills to Prepare for the AI Era: Beyond Computational Thinking 14
5. Why We Need to Collaborate with AI: Education Beyond Technology: Moving to Partnership 15
6. Values Education, Rather Than Ethics: A New Educational Focus Regulating the Direction of Technology 18
7. The Need for a New Learning Theory for an AI-Based Society: A Shift in the Education Paradigm Based on Human-AI Relationships 19
8. Education that focuses on possibility rather than fear: Humans are flexible in the face of technological change and ultimately adapt to their environment. 20

03. Structure of the AI Education 2.0 Model 22

1. AI Education 2.0 Model 23
2. Definition of detailed model terms 24

04. Details of the AI Education 2.0 Model 26

1. Why does AI basic social education require a new learning theory? 27
2. What is digitalization? 29
3. What is AI thinking? 38
4. Learners' cognitive level improved through AI thinking 46
5. AI Understanding Education: What is AI and how does it work? 49
6. AI Collaborative Education: What, How, and Together Can We Do with AI? 52
7. AI Value Education: How can humans and AI advance toward a valuable society? 58

References 61

01

Introducing AI Education 2.0

Machine Translated by Google

1 Why 'AI Education 2.0'?

In response to the transition to an 'AI-based society' that has arrived with the rapid development of artificial intelligence technology, new educational programs are also being introduced in the field of talent development.

A paradigm is required. This does not stop at the level of simple acquisition or use of technology, but rather requires all citizens to live with AI in a society.

This is a shift towards education that cultivates the basic capabilities necessary for human development. In particular, it is necessary to ensure that artificial intelligence can reach a level of cognitive ability similar to that of humans.

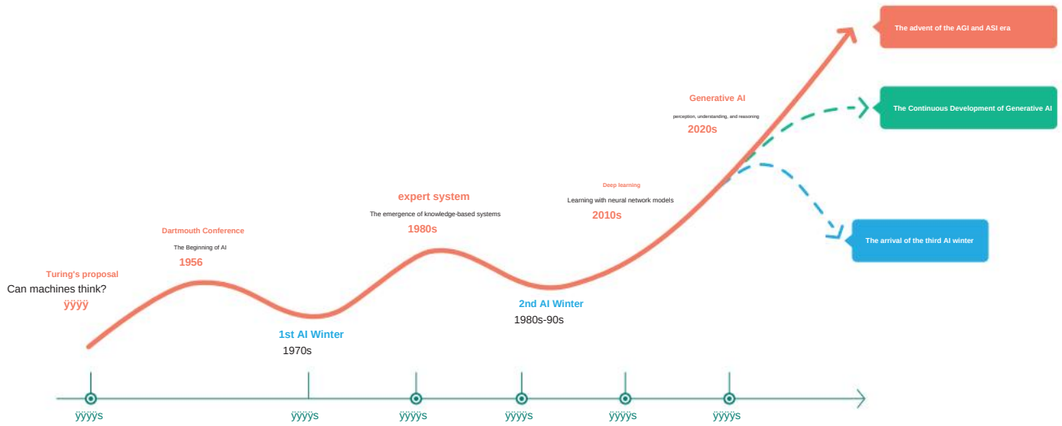
With the era of AGI (Artificial Intelligence) in mind, we propose 'AI Education 2.0' as a practical educational model suitable for it.

This educational model goes one step further than the existing AI education that mainly focused on 'using AI as a tool' or 'integrating it with other subjects.'

Furthermore, it is based on a transformative mindset that recognizes AI as an "object of cooperation." In other words, AI is not simply a user, but rather,

The core characteristic of 'AI Education 2.0' is to make AI an educational partner that solves problems and creates value together with humans.

A shift in perspective is a core foundation for building an AI-based society, and the education for this is 'AI Education 2.0.'



[Figure] Scenario after the 3rd AI Boom

Accordingly, the capabilities and underlying theories pursued by 'AI Education 2.0' are as follows.

- The goal is to provide education that fundamentally understands artificial intelligence technology based on digitacy and integrates the core values of an AI-based society.
- Develop AI thinking skills that go beyond computational thinking that utilizes computing power and collaborate with AI to solve problems.
- By adopting meta-collaborativism as a learning theory, we expand the collaboration between humans and AI and present a new learning structure.
- Symbiotism is a next-generation learning theory that assumes the post-AGI era, and goes beyond meta-cooperationism to pursue human-AI co-evolution.

Explore the direction.

References

From DX to AX

Digital transformation (DX) has focused on digitizing and automating existing analog or IT-based processes. However, AI technology is now demanding an evolution beyond digitized processes toward AI transformation (AX), where processes become intelligent and autonomous. This trend is clearly reflected in Microsoft CEO Satya Nadella's definition of "DX is automation, and AX is autonomy."

AX aims to go beyond simple task automation to enable AI-based autonomous decision-making and execution across all corporate operations. Ultimately, AX The shift to AI is not simply a technological innovation. It's a process by which companies are transforming into entities capable of making and executing decisions on their own through AI. This shift is driving a fundamental restructuring of work methods, value judgments, and human-technology relationships, extending beyond organizations to society as a whole. Corporate change is accelerating societal change, and the AI transition is ushering in a new way of life for all citizens.

division	DX (Digital Transformation)	AX (AI Transformation)
target	Analog ÿ Digital Conversion, Automation-centric	Beyond digitalization, 'intelligence and autonomy' AI-based decision-making and execution
Key technologies	System automation (RPA, etc.), ERP, Cloud	LLM, AI Agent, Autonomous judgment systems, predictive AI
Decision-making level, rule-based, automated execution		AI recognizes, analyzes, and judges the situation to make the optimal choice.
applied area	Part or all of the business process Automation	All levels, from field operations to mid-level decision-making and top strategy formulation

Reference: https://enterprise.kt.com/bt/dxstory/2959.do

EX (Education Transformation) - A New Look at Education

In the field of education, digital transformation, centered on edutech and powered by digital technology, has contributed to increasing the efficiency and accessibility of education. However, as we enter the era of AX, where AI intelligentizes and automates society as a whole, education also requires a fundamental transformation. In other words, we need to establish a direction toward EX (Education Transformation). EX is not about teaching how to effectively use AI technology, but rather about how to coexist with AI. This is an educational innovation that requires not simply the acquisition of technology, but a shift in how we exist, relate, and judge. It can be seen as a process of cultivating a new kind of citizenship in the AI era.

Conversion factor	Existing education (DX-focused)	EX: AI Education 2.0-based education
Object of Learning: Humans vs. Technology		Human + AI (Symbiotic Partnership)
Digital Utilization Competency as a Goal of Education		Understanding AI ÿ Collaborating ÿ Putting Values into Practice
Educational method: correct answer-centered, content-infused		Question-based, relationship-centered, meaning-creating
The role of technology as a tool (automation)		co-agency, augmentation
Learner-oriented knowledge consumer		Reflective explorer, resonant collaborator, creative practitioner

Machine Translated by Google

2. Goals and Targets of 'AI Education 2.0'

target

The core goal of AI Education 2.0 is to cultivate citizens' capabilities to realize an AI-based society. To achieve this, it has the following goals:

ÿ Ability to accurately understand AI: Understand the concepts, principles, and limitations of artificial intelligence in a balanced way within a technological and social context.

ÿ Practical skills to collaborate with AI: Recognize AI as a partner for collaboration, not a tool, and be able to perform tasks of human-machine collaboration.

Have the power and ability to execute.

ÿ Value sensitivity required in the AI era: Values that enable ethical judgment and social responsibility centered on values in a society where humans and AI coexist

Form a directional attitude.

ÿ A new framework that goes beyond conventional computational thinking: A step beyond automation-centered computational thinking, AI's judgment and learning structure

We look at and solve problems based on AI thinking that reflects this.

The type of talent we pursue

The type of talent that AI Education 2.0 aims for goes beyond simple technology utilization skills and has the in-depth knowledge required in an era where we live with artificial intelligence.

This education aims to develop responsible future citizens equipped with critical thinking and community awareness. The talents this education seeks are:

It is formed around characteristics.

First, they are reflective inquirers. These individuals go beyond simply receiving information and ask fundamental questions about humanity, technology, and society, exploring the depths of their understanding.

We pursue understanding. We continuously reflect on our own experiences and thoughts, and we develop a critical and philosophical perspective on AI technology and social phenomena.

Approach it as such. By exploring the essence of the problem through the fundamental question of 'why', and based on inner growth and essential understanding, it is possible to live and learn.

Expand your insight into Korea.

Second, they are creative collaborators. These individuals empathize with and creatively communicate with AI and humans, and with each other.

They are beings with the ability to collaborate. Beyond simple division of roles, they understand and coordinate each other's differences and collaborate to create new ideas.

We demonstrate collaborative capabilities. We consciously collaborate around shared goals and values, and we coordinate relationships with others, technology, and society.

It leads to a relationship of continuous mutual development.

Third, they are responsible practitioners. These individuals do not remain technology-centric, but rather focus on human-centered thinking within the social context connected by AI.

They are beings who practice values and ethics. They don't just solve problems; they also understand the meaning of the process and results of those solutions to society.

We reflect and act. We connect and disseminate the value we create with the community, recognize our responsibility in utilizing technology, and work toward a sustainable future.

Maintain a practical attitude.

Ideal talent	Key Concepts	Competency
reflective seeker	Based on deep reflection on technology, humanity, and society A person who understands the essence and seeks a better direction through continuous exploration	ÿ Reflective thinking skills: Reflecting on one's own experiences, knowledge, and emotions The ability to see into the essence of a problem and derive new perspectives ÿ Essential inquiry skills: Core principles and Understand the structure and expand your understanding by asking fundamental questions. Ability to do
creative collaborator	Collaborating with AI, people and the environment to create new ideas and A person who creatively solves problems through perspective	ÿ Creative collaboration capabilities: By integrating various resources, people, and AI perspectives The ability to collaborate and solve problems in new ways ÿ Empathetic communication skills: Understand others' emotions and perspectives and maintain an open attitude The ability to communicate and coordinate
Responsible practitioner	Human-centered values in a society connected to AI Respect and ethical judgment and practice based on it Talents who contribute to the community	ÿ Value judgment ability: Interpret situations from a social and moral perspective The ability to determine the right direction ÿ Ethical practice capacity: Responsibly following the values and judgments decided The ability to act and consider the consequences

Target

AI Education 2.0 is a program that requires understanding and practical skills in AI throughout the entire life cycle in preparation for a future society where we will live with artificial intelligence.

It targets all citizens. Specifically, the scope of application is defined by four educational stages:

ÿ Primary and secondary education (K-12): To develop a basic understanding and desirable attitude toward AI for students who will become the main players in future society.

This period is the starting point for naturally acquiring the basic digital skills and AI collaboration skills necessary to understand AI.

ÿ Higher education: The younger generation moving towards professionalism will critically understand the structure and limitations of AI technology and various academic and social issues.

We help you leverage AI to solve problems. This process emphasizes collaborative problem-solving and value judgment skills.

ÿ Vocational training: For those who need practical collaboration with AI in industrial and work environments, it provides structural understanding and practical application of AI systems.

Develop your capabilities and strengthen your ability to utilize and adapt to AI in your specific role. Challenge the limits of AI technology to transform the world and shape the future with innovative capabilities.

This is emphasized.

ÿ Lifelong education: Continuously responding to technological changes, ensuring that all generations lead self-directed lives and fulfill social responsibilities even amidst the changes of the AI era.

We support this. Specifically, we aim to prevent digital exclusion and ensure that all citizens have basic rights and understanding of AI.

Machine Translated by Google

02

AI Education 2.0

Core principles of the model

1. The foundation of the AI Education 2.0 model: It must start with digital.

The starting point of AI Education 2.0 is not simply dealing with artificial intelligence technology, but understanding how that technology was created. The AI technology we use today is superficially composed of complex concepts such as data, algorithms, machine learning, and deep learning, but At the very bottom lies the digital world of 0s and 1s, the operating principles of digital logic.

The foundation of AI technology is mathematics and computer science. While mathematics enabled theoretical structures and modeling, computer science translates these into logical commands. AI and machine language were embodied to create today's digital environment. All AI technologies ultimately operate as computing systems that operate on digital computation. , and without understanding this, it is difficult to understand how AI works and its limitations.

Digital is not simply a signal of 0 and 1. It is a way of thinking that quantifies the world, processes it logically, and abstracts it. It can be said to be the language of modern technological civilization. Computer science has invented various algorithms and structures based on the concept of digital, and it is It again became the basis for artificial intelligence and generative models.

Today, we live in an era where AI writes code on its own and produces creative results, but all of that calculation and judgment is ultimately digital. It's built on expression and processing. Without an understanding of digital, it's impossible to directly address AI, like building a building without a foundation. It's work.

Therefore, AI Education 2.0 must go beyond simply "education that utilizes AI" and begin with an understanding of what AI stands for. Ultimately, Education that properly understands, utilizes, and evaluates AI's value must begin with a deep reflection on digital.

It's digital

by Nicholas Negroponte 1995

The core of the digital age is the transition of the world's building blocks from 'atoms' to 'bits.' Bits have no weight or shape. However, it offers tremendous efficiency and freedom in the storage, replication, and transmission of information. This has led to a rapid transition from physical to digital forms of all media—books, newspapers, music, videos—and a restructuring of human lifestyles and communication methods centered around bits.

Negroponte explains that the digital society is changing in four directions: decentralization, globalization, harmonization, and decentralization. Information is centralized. A digital global village where individuals are directly connected through networks rather than centralized networks, and the entire world is connected beyond regional and national boundaries. At the same time, individual voices are strengthened and society changes into one where diverse perspectives coexist.

Digital technology is also changing the way humans interact with computers. Negroponte believes that a good digital interface should not be a simple tool, but a "digital agent" that understands and responds to humans. Technologies such as voice recognition, emotion recognition, and personalized information provision are transforming human life. Computers are evolving in a direction where they are coordinated around .

One of his signature predictions was the "Daily Me" concept, a personalized information delivery system tailored to the user's tastes and interests. This could be considered the prototype of modern algorithm-based news feeds and recommendation systems. He also emphasized the structural restructuring of digital infrastructure, advocating for a transition between wireless and wired communication structures through the "Negroponte Switch."

While Negroponte is optimistic that digital technology will egalitarianize society and empower individuals, he also urges caution regarding new ethical issues such as copyright, privacy, and the information gap. A digital society is not complete with technology alone, and it requires a corresponding approach. His consistent message is that systems, philosophy, and human-centered values must be designed together.

Machine Translated by Google

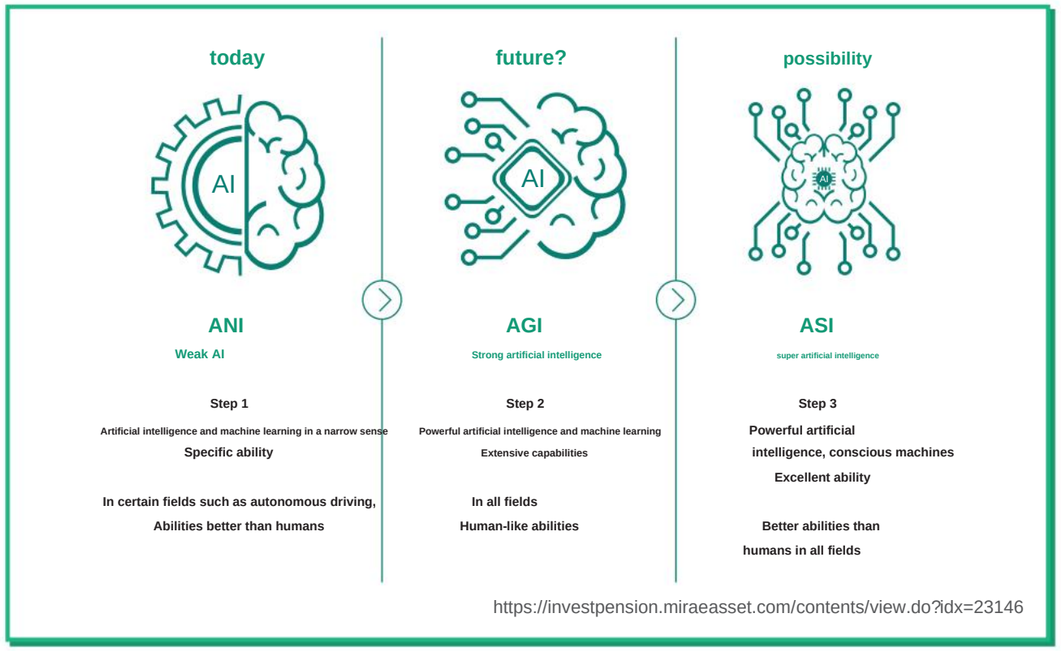
2. The singularity of education considering AGI and ASI: Education designed with the future of technology in mind.

AI Education 2.0 doesn't focus on the fragmented functions or performance of existing AI technology. It simply aims to provide education that keeps pace with current technological advancements. Rather, it aims to build a new educational paradigm based on the future possibilities that AI technology can achieve.

The history of artificial intelligence began with Alan Turing's abstract computing machine, and has gone through the first, second, and third AI booms and AI winters to reach the present. We can lay the foundation by understanding AI technologies of the past and present, but if education remains limited to the past and present, the pace of technology will be slow. We are stuck in a reactive education system that cannot keep up with the times. AI Education 2.0 aims for AI-centered education during this period of great transformation. This is not just about 'AI Beyond 'education utilizing AI', it means education that reconstructs human thinking, life, and society with AI. Therefore, AI education is a technology that is based on reality. While reflecting on this, we must also proactively consider future technologies.

That future is coming. The advent of AGI (Artificial General Intelligence) is no longer a fantasy. It will think, judge, and act like humans. AGI, applicable to various fields, will drastically change the structure of future society, labor, creativity, and values. Education will play a key role in this change. Rather than passively responding, it should become the foundation for actively preparing humans to live in the AGI era.

Furthermore, considering the possibility of artificial intelligence (ASI), education requires even more fundamental questions and innovative imagination. It's not just about technology. It should be an education that allows us to coexist with technology, take responsibility, and expand our humanity, rather than learning AI. AI Education 2.0 is a predictable In the flow toward technological singularity, students will develop the thinking skills, ethics, and ability to proactively design and respond to future changes. It should be an education that fosters social sensitivity.



https://investpension.miraeasset.com/contents/view.do?idx=23146

- ANI (Weak Artificial Intelligence): AI systems designed and trained for specific tasks or a narrow range of tasks, such as playing chess or recognizing speech. It excels at performing predefined tasks such as:
- AGI (Strong Artificial Intelligence): An AI system that can understand, learn, and apply knowledge across a wide range of tasks, similar to human intelligence. AGI possesses cognitive abilities similar to humans.
- ASI (Artificial Superintelligence): An AI system that surpasses human intelligence and capabilities in all aspects. ASI possesses cognitive abilities far surpassing those of humans. and could potentially lead to groundbreaking advancements or pose existential risks.

3. Why we need to know AI 'properly' above all else: Proper cooperation and value come from understanding

Today, AI is already deeply embedded in our daily lives. Those who can utilize and apply AI are at an advantage in economic opportunities and technological advancement. The era of leading and having social influence has arrived. In this context, "Let's make good use of AI" and "Let's achieve innovation through AI" The discourse is strongly established.

But this raises a fundamental question: Is true utilization possible without a proper understanding of AI? The term "tool" implies that technology is treated as a tool, but using it without knowing the principles and operation of the tool is superficial. It can be limited to passive use, relying on functionality. True utilization involves understanding the limitations of the tool, judging it according to context, and exploring new approaches. It must be transformed or creatively reconstructed. To this end, the structure, operating principles, judgment method, and algorithmic nature of AI must be considered. An understanding of bias must come first.

Furthermore, value judgment and ethical reflection, which are considered core competencies in the AI era, are empty slogans without sufficient knowledge and understanding of AI. If you don't even know why AI produced a certain result or what data and structure it learned from, you can't criticize the result. It's also impossible to suggest directions for improvement. In other words, ethics and values are based on understanding.

All knowledge can only be said to be "known" if one understands its concepts and principles. If we emphasize only "utilization" of AI without "knowing" it, it will lead to a future society that is incomplete. There is a risk that this will lead to an attitude of uncritical acceptance and consumption, treating core technologies as if they were a 'black box'.

Based on this awareness of the problem, AI Education 2.0 aims to establish the starting point of all education, including cooperation with AI, value judgment, and ethical practice, as 'in-depth understanding of AI. It is set to 'understand'. Students should not simply perceive AI as a convenient technology, but rather explore its internal structure, limitations, and possibilities. You will grow into an active being who can analyze yourself and utilize or redesign it for the better.

As citizens of the AI era, the most fundamental starting point for education should be the intelligence to properly understand and handle technology.

Machine Translated by Google

When we stop understanding the world

"When We Stop Understanding the World" illuminates the anguish and ethical questions behind scientists' groundbreaking achievements. Fritz Haber He developed fertilizer to save humanity from starvation, but at the same time he also created poison gas to be used for mass murder. Heisenberg and Schwarzschild And Grothendieck, while reaching the pinnacle of brilliant theory, came at the price of mental isolation and the brink of madness. This work is a scientific He warns that technological advancement does not necessarily mean progress, and that failing to question its principles, purpose, and reason can lead to destruction.

Stories like this pose the same questions for us living in the AI era. We are currently "utilizing" numerous technologies under the banner of AI. But how does this technology learn and make decisions? What data does it rely on? What biases and distortions does it introduce? How much do you understand about what could happen?

AI is never neutral, and human intentions and choices are fully reflected in its code and algorithms. Nevertheless, viewing AI as merely a tool for convenience, passively using it while neglecting its principles like a "black box," risks repeating the ethical failures of past science.

The core message of "When We Stop Understanding the World" is simple: "When we stop understanding, responsibility stops." Technology is a tool for human understanding. Only through reflection can we have true value and direction. The same goes for AI. If we do not make an effort to properly understand AI, If so, we may be reduced to beings who cannot criticize or take responsibility for the results of that technology.

Therefore, education in the AI era must go beyond simply nurturing "technology-savvy individuals." It must explore the other side of technology and the relationship between humans and technology. We must cultivate intelligent and ethical citizens who can philosophically reflect on this and design "proper cooperation" with AI. This is truly understanding technology "properly" and using it "well."

4 New Thinking Skills for Preparing for the AI Era: Beyond Computational Thinking

Humanity has redefining the direction and purpose of education with each new era, as it encounters new technologies. The school system of the Industrial Revolution was designed to meet the needs of industry. The school was designed to cultivate talent, and the classroom was a space that embodied that purpose. Each subject was systematized according to the needs of industrial society. It has become a 4C competency (critical thinking, creativity, communication, collaboration) to cultivate a well-rounded human being with a balance of knowledge, skills, and attitudes. This was emphasized.

The digital revolution and the advent of the information society have presented new challenges to education. The ability to understand and utilize digital technology is essential. It has become an essential skill for survival, and computational thinking has emerged as a new core competency in education. This is the ability to think in terms of structuring problems, algorithmizing them, and solving them through computer-based automation, and is essential for adapting to the digital world. It supported my practical ability.

But now we are facing a more fundamental shift. AI is no longer a mere tool; it is becoming more like human thought. Machines are evolving into 'intelligent beings' that can think, judge, and create. Machines are becoming capable of thinking and making choices in a way similar to human mental capabilities.

As AI becomes more prevalent, the need for complex and critical thinking emerges, a need that cannot be addressed solely through simple technological use or procedural problem-solving. The talent of the era is not just someone who can use digital technology, but also someone who can think, collaborate, create with AI, and think about the results. You must be a being capable of reflection.

This requires AI thinking that goes beyond conventional computational thinking. AI thinking is not simply logical or procedural. Beyond thinking, understand AI's judgment methods and operating principles, critically analyze the results, and collaborate with humans and AI to solve problems. It is the ability to think and create new values.

AI Education 2.0 aims to revolutionize the thinking skills of new talents who will thrive in a future where humans and AI coexist at a turning point in their history. And at its core lies the ability to expand thinking itself through AI, or AI thinking.

Computational Thinking

Computational thinking is a way of thinking that logically and systematically solves complex problems based on the principles of computer science. It is not simply a computer It is not the technology that is used, but the ability to understand a problem, design a solution, and automate the process.

The main elements are:

ÿ Decomposition: Dividing a complex problem into smaller problems.

ÿ Pattern Recognition: Finding recurring rules or similarities in a problem

ÿ Abstraction: Removing unnecessary information, leaving only the essentials.

ÿ Algorithm Design: Creating a clear procedure for problem solving

ÿ Automation: Implementing code to enable machines to automatically solve problems.

Computational thinking is the ability to analyze problems, extract key points, design procedures, and automate solutions in the digital age. It is a core competency required for everyone living in this world.

5 Reasons to Partner with AI: Moving Education Beyond Technology to Partnership

Today, we frequently encounter scenes in our daily lives where AI surpasses human capabilities. It's already been used in areas like Go, law, writing, art, and medical diagnosis. In various fields, AI is demonstrating performance that surpasses that of human experts.

So, we must ask: How far will AI advance?

The technological singularity proposed by Ray Kurzweil is no longer a story of the distant future. AGI (artificial intelligence), and even further At a time when the possibility of ASI (Artificial Superintelligence) is becoming increasingly realistic, we cannot simply remain at the level of 'using AI as a tool'. We need to consider whether to establish a new relationship with AI or to collaborate with it.

Machine Translated by Google

So what is cooperation?

Traditionally, cooperation was an activity in which people formed relationships based on mutual understanding, responsibility, and division of labor for a common goal.

However, AI is not human. It lacks emotions, intentions, and subjectivity, so it cannot be held to the same standards as human cooperation.

However, AI has become a force that can solve problems together, influence decision-making processes, and contribute substantially to creation and production.

In this respect, collaboration with AI can be defined as a new type of 'extended collaboration' or 'heterogeneous collaboration', and education should be tailored accordingly.

It needs to be redesigned.

Why do we still struggle to accept AI as a partner? The reasons are varied.

Collaborating with AI may be driven by fear of technology, threats to human self-esteem, or a lack of clarity about who is responsible.

What's important is that we're already collaborating with AI. AI that automates coding, generative AI that co-creates content, and business AI.

AI assistants that assist us are all early forms of practical cooperation.

If we recognize AI as a collaborative partner, what changes will be necessary in education?

Beyond simply acquiring skills, we must learn to understand AI's judgment, share roles, and achieve common goals.

Rather than simply using AI as a tool, we coordinate the flow of thought with AI, exchange feedback, and jointly produce creative results.

It is an experience.

This training has the following implications:

Fostering Relational Sensitivity Between Humans and AI

ÿ Redefining roles, responsibilities, and ethics

ÿ Strengthening critical acceptance and restructuring capabilities for AI results

ÿ Changing learning design for co-creation, co-exploration, and co-problem solving

To this end, education can introduce the following methods:

ÿ Project-based learning with AI

ÿ A metacognitive-focused class that critiques and redesigns AI results.

ÿ Practical activities based on scenarios of collaboration between AI tools and humans

ÿ Human-AI joint judgment training through role-playing and simulation

The ability to collaborate with AI goes beyond simple technological use; it's a core competency that fosters new forms of thinking, judgment, responsibility, and sensitivity.

AI Education 2.0 must clearly recognize this and shift education toward fostering "humans who can collaborate with AI."

The meaning and prerequisites of a partner

1. Basic prerequisites for cooperative partners

ÿ Shared goal: Cooperation is only possible when partners have a common purpose and vision. If the goals are not aligned, cooperation is simply ineffective.
It may end up being a division of roles or a transaction.

ÿ Mutual trust: There must be a basic trust that the other party will be trusted, respected, and keep promises. If trust is broken,
Cooperation turns into conflict or control.

ÿ Equality in partnership: Cooperation is not a command-and-obedience relationship, but rather presupposes horizontal interaction. The other party's expertise or
Experiences must be recognized and given equal opportunity to contribute.

ÿ Autonomy and responsibility: Each person must be able to autonomously perform his or her own role and responsibilities and take responsibility for them.

2. Capabilities and attitudes required of cooperative partners

ÿ Communication skills: The ability to clearly express one's thoughts, listen to the opinions of others, and avoid conflicts or misunderstandings.
A transparent and open communication attitude for

ÿ Empathy & Respect: An attitude that understands and cares for each other's position, and an attitude that acknowledges differences and respects diversity.

ÿ Negotiation & Coordination: The ability to find common ground and coordinate when interests or opinions differ, and the ability to flexibly adjust without harming the group's goals.

ÿ Co-problem solving: The ability to analyze problems and derive solutions by pooling each person's capabilities, and the ability to share ideas and create possible solutions.

ÿ Continuous Feedback & Reflection: An attitude to improve by giving and receiving mutual feedback during the collaborative process, not only the results but also the
The ability to look back on the process together is also necessary.

meta-cooperation

"Meta-collaboration" refers to a higher level of cooperation than general cooperation or collaboration. Here, "meta"
is a prefix meaning 'transcendent', 'superior', 'of onesell'.

1. Higher-level collaboration: This is not simply collaboration, but collaboration that designs and coordinates the collaborative structure between different collaborating entities.
Examples: Coordinating relationships between individuals and AI, groups and AI, institutions and technological systems.

2. Collaboration on Collaboration: This is a process of jointly reflecting on and improving the methods of collaboration itself, discussing and implementing ways to better collaborate. For example, in an educational setting, co-designing how teachers, students, and AI can share roles and interact effectively.

Human-to-human cooperation vs. human-to-AI cooperation (a meta-collaboration perspective)

division	human-to-human cooperation	Human-AI collaboration
	Cooperative entities are autonomous beings with emotions, consciousness, and values.	Algorithm-based non-autonomous and non-conscious beings
	Prerequisites: mutual understanding, empathy, and trust	Clear role definition, distribution of responsibilities, and data-driven trust
	Sharing communication styles, including verbal, nonverbal, and emotional context	Explicit protocol, logical instructions, interface-based
	Designing a collaborative structure that allows for implicit agreements or flexible adjustments	Requires predefined rules or human-driven coordination
	Problem-solving methods: collaborative exploration, creative coordination, and negotiation	Rule-based optimization, tailored to human objectives
	Meta-collaboration potential: AI can creatively discuss and reconfigure the collaborative structure itself. AI does not participate in the structural design; humans are responsible for coordination.	
	Emphasize moral responsibility according to the ethical relationship and context of cooperation.	Human ethical design and control are key

Machine Translated by Google

6. Values Education, Rather Than Ethics: A New Educational Focus Regulating the Direction of Technology

Ethics is an essential topic in education in the AI era. Now, as AI's judgment, creation, and operation deeply impact human lives, we must consider the implications of ethics. Naturally, one might ask whether AI can be "ethical," or what ethics the humans who create AI should follow. But here, We need to think about it one step further.

Can we demand ethics from technology?

And is it really possible to shift all ethical responsibility onto developers or designers?

Ethics are, in principle, standards for distinguishing between good and evil, right and wrong, but they themselves change according to the times, culture, and social consensus. It is a relative and context-dependent concept. Human ethics also does not have absolute standards, but is based on human philosophy, experience, and values. It depends. Attempts to apply 'ethics' to AI technology indiscriminately sometimes hinder technological development or obscure responsibility. It also creates. Therefore, AI Education 2.0 proposes a shift beyond the traditional "ethics-centered education" to "value-centered education."

If ethics asks about the legitimacy of an action, 'value' is an active question about what to pursue and why we should choose that direction. Ultimately, it is humans who create and use technology, and the ultimate direction of technology is determined by the "values" pursued by humans. For example, Even with the same AI technology, it can look completely different depending on which values are prioritized, such as life protection, fairness, autonomy, community, and sustainability. can be implemented as

The education pursued by AI Education 2.0 is as follows.

ÿ Rather than defining the ethical responsibility of AI technology with a single standard, foster the ability to discern and discuss diverse value standards.

ÿ Education that cultivates the habit of making value judgments about 'what this technology will be used for' before using the technology.

ÿ Education in value judgment that can lead to community consensus and human-centered decisions even in complex and contradictory problem situations.

ÿ Fostering citizens who have a balanced understanding of the limitations and possibilities of technology and who can select and design technology in a way that benefits humanity.

If AI ethics education was a defensive approach to reduce damage caused by technology, AI values education is a way to improve human and society through technology. It's a proactive approach to moving forward. AI education now goes beyond asking "what is right?" to "what should we live for?" It must evolve into an education that asks and thinks together.

The difference between ethics and values

1. Difference in definition

ÿ Ethics: The norms or standards for judging right and wrong, good and evil in a society or group. "What is morally desirable?"

It presents criteria for judgment.

ÿ Value: The importance or meaning of an object, action, or belief, that is, what is considered important. What an individual or group considers important

It is a judgment on "Do you consider it more important?"

2. Differences in perspective

ÿ Ethics are standards for what one should do. Ex) "You should not lie." (Social norms)

ÿ Values are judgments about what is 'important' e.g. "Honesty is important." (personal beliefs)

3. Differences in application

ÿ Ethics are standards of conduct that are usually required from outside, such as society or a group, and must be observed within the community.

ÿ Values are formed within an individual or society and are the criteria for choosing what to prioritize.

In other words, ethics is the standard of 'right and wrong', and values are the judgment of 'importance'.

7. The Need for a New Learning Theory for an AI-Based Society: A Shift in the Education Paradigm Based on Human-AI Relationships

The education that humanity has pursued has always been closely intertwined with technological advancements. The mass education system of the Industrial Revolution was characterized by machines and factories. The school system was created to support the central society, and since then, up to the information society, education has kept pace with the changes of the times and technology. It has been constantly changing its form as it adapts.

Technology has advanced rapidly, and education has been pursuing and embracing it, seeking new directions. Along with this, understanding of learners and The methods of education have also evolved. Behaviorist learning theory viewed learning centered on external stimuli and rewards, while cognitivism viewed learning as internal to humans. Constructivism emphasized the information processing process, while learners are the ones who construct meaning. These three major theories All of them were learning theories based on the premise of 'interaction between humans.' In other words, they focused on the relationship between teachers and learners, or between learners. These are theories.

But now, we have entered an era where humans are no longer the only learning partners. AI is no longer a mere information provider or subject of education.

Machine Translated by Google

Rather, AI is emerging as a subject and collaborative partner in learning. AI responds to learners' questions in real time, analyzes the learning flow, and It is replacing or supplementing some of the roles of the instructor, such as suggesting new alternatives. Furthermore, it is solving problems and creating together with AI. Ability is now becoming a core competency for survival in the future society.

These changes cannot be sufficiently explained by existing learning theories alone, and it is necessary to pursue their essence in education and support students' growth. It has limitations because the theory so far has been an educational structure based on the premise of 'human-human relationships'. AI is an inhumane But now that intelligent beings have emerged as educational partners, we need a new educational theory that centers on human-AI interaction.

AI Education 2.0 aims to achieve this by embodying "meta-collaborativism" and "resonance," where humans and AI interact and evolve together. It suggests the need and direction for new learning theories such as (Symbiotic Learning). These theories expand the existing teaching-learning relationship. Thus, it constitutes a multidimensional learning structure that includes human-AI-environment interaction, metacognition and emotional connection, value judgment and co-creation. I want to do it.

In the future, we must consider not only AGI (Artificial General Intelligence), but also ASI (Artificial Superintelligence). This will affect the target and method of education. This raises the need to completely redesign assessment and objectives themselves. In other words, education is not just for humans, but for humans and AI to grow together. We must move toward education that fosters individuals who can actively understand and responsibly utilize AI.

8. Education that focuses on possibility rather than fear:

Humans are flexible to technological changes and ultimately adapt to their environment.

There is a recurring narrative whenever technology advances radically.

"AI will replace all human jobs."

"New technologies will make human roles irrelevant."

These discourses often provoke anxiety, fear, and resistance, spreading technological skepticism throughout society. However, looking back at history, we see that We come to see a completely different truth: humanity has always survived technological changes, and has even used them as a springboard for growth. When machines first appeared, human physical labor was threatened, when computers appeared, there was talk of a crisis in office work, and when the Internet spread, When this happened, there were voices of concern about the collapse of society.

But throughout all these ages, humans have constantly evolved, working, learning, and creating in new ways. Technology has replaced humans. Rather than being taken away, it has served as an opportunity for humans to reconfigure and expand their capabilities. Of course, change is uncomfortable, unfamiliar, and sometimes painful. However, humans are inherently learning, adapting, restructuring and responding to problems.

These characteristics lead not only to simple survival, but also to the ability to create new lifestyles and values. The same holds true in the era of AI technology. This change is of a different level of speed and impact than before, but human adaptability and resilience are no less. We are now in a state of constant competition with machines. We are entering an era of cooperation and resonance with machines, expanding our humanity, rather than an era of competition.

AI Education 2.0 begins from this perspective. Instead of emphasizing the fear of technology, it emphasizes human resilience, flexible thinking, and creative restructuring. Trust in your abilities. Rather than instilling a vague fear of change in students, you should proactively understand the change and develop your own abilities. The role of education is to connect with life and foster the ability to actively collaborate with AI to find solutions to problems.

Such training should include:

- Critical understanding of technological change and the ability to interpret it subjectively
- The ability to flexibly respond to change and create new environments on one's own
- Creative integration ability that can merge unique human emotions, imagination, morality, and community with technology
- Continuous learning that allows you to constantly adapt to new challenges without fear of failure.

We must design education based on trust in people. Technology is not meant to replace humans, but to enhance human potential. It's a tool for expansion. AI Education 2.0 aims to cultivate human beings who are unafraid of technology, flexibly adapt to change, and proactively design their own lives. We believe in this and want to cultivate such talents.

Machine Translated by Google

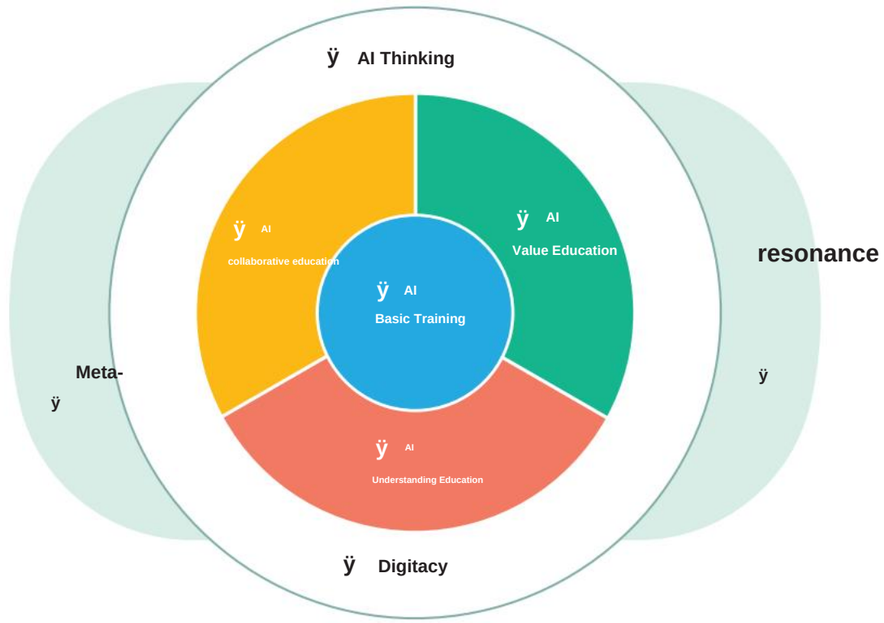
03

Structure of the AI Education 2.0 Model

1 AI Education 2.0 Model

The AI Education 2.0 model structured the core competencies that citizens must possess in an AI-based society into three educational areas, and based on this It is an integrated model that visually expresses the educational transformation in the AI era.

AI Education 2.0 Model



At the center of the model is 'AI basic education', which is the education that becomes the center of a society where AI is the basis, and this means that all learners can become AI-based in the society. This concept integrates the essential educational elements necessary for survival. Basic education includes AI understanding education, AI collaboration education, and AI value education. It consists of three areas, each with complementary yet independent educational objectives.

ÿ AI understanding education builds a knowledge foundation based on a critical understanding of the principles and operating structure of AI technology.

ÿ AI collaborative education goes beyond the coexistence of humans and AI, fostering the relationship skills to solve problems by interacting with AI as a collaborative entity.

ÿ AI value education aims to cultivate civic sensibility and ethical judgment that can practice human-centered values even amid technological advancements.

Aim for.

These three educational areas are integrated into AI Thinking, a core thinking ability in the AI era, which goes beyond simple computational thinking. It refers to a high-level thinking system that includes learning structure, judgment method, and problem solving based on human-AI collaboration.

Machine Translated by Google

The outer rim surrounding the core and mantle of the model emphasizes digitacy as the foundation of AI thinking. This includes ICT, software,

This is a fundamental competency that allows you to proactively adapt and creatively respond to the digital environment based on basic knowledge of data and AI.

In addition, the two philosophical learning theories that support the entire AI education model are meta-cooperationism and resonanceism, which have developed into existing learning theories.

It is placed on a foundation.

Meta-collaborativeism emphasizes that education should be reconstructed together with technology through a structure of multi-layered cooperation between humans, AI, and society, and resonance theory

It aims for an ontological transformation of education through a relationship in which humans and AI interact and evolve together.

This model goes beyond AI education that simply teaches technology, and aims to educate people who will live with AI and become the basic human beings in an AI-based society.

It contains a practical educational philosophy that fosters the ability to exist meaningfully as a citizen.

2 Model diagram detailed term definitions

Terminology	meaning
ÿ Basic AI training	This is an education program to develop the basic capabilities that all citizens living in the AI era must have, including understanding, cooperation, and value of AI. Core-based education encompassing three areas
ÿ AI Understanding Education	Understand what AI is and how it works, and be able to critically analyze the principles of algorithms, data, and artificial intelligence systems. Education that develops existing abilities
ÿ AI collaborative education	Recognize AI as a collaborative partner, not just a tool, and develop the ability for humans and AI to work together to solve problems and perform tasks. Education that nurtures
ÿ AI value education	Understanding the impact of AI technology on society and the importance of human-centered values, based on coexistence, responsibility, ethics, and community awareness Education that develops the capacity to act
ÿ The ability to critically adapt to the digital environment and creatively solve problems based on digital ICT, SW, data, and AI	
ÿ AI thinking ability	This is a higher-order thinking approach that considers AI's judgment and learning methods, extending beyond traditional computational thinking. It encompasses the ability to recognize problems, reframe them from an AI perspective, and explore solutions based on human-AI collaboration.
ÿ Meta-cooperationism	It is a theory that focuses on continuous and mutually evolving collaborative learning between humans and artificial intelligence (AI), where learners develop metacognitive skills. Through reflection and meta-thinking, we can adjust and restructure our thinking in relation to AI, and acquire knowledge and problem solving skills. A learning paradigm that co-constructs the process
ÿ Resonance	An ontological perspective that emphasizes the relationship between humans and AI that goes beyond mere coexistence and mutual interaction and co-evolution. An educational philosophy that involves transformation

*Sensitivity: The process by which two entities (or systems) react sensitively to each other's changes or states, and create new meanings or states together through non-unidirectional interaction.

References

AI Education 1.0 Education Model

In 2020, the Korean Society for AI Education designed and presented a model for AI education as follows. At the most basic level, basic education is provided to understand the operating principles of humans and AI,

including mathematics, science, and computer science. This is to develop logical thinking skills and a conceptual foundation for AI.

This is the starting point for forming a foundation. Based on this foundation, learners will develop a critical perspective on technology through AI education that understands the concept of artificial intelligence, how it works,

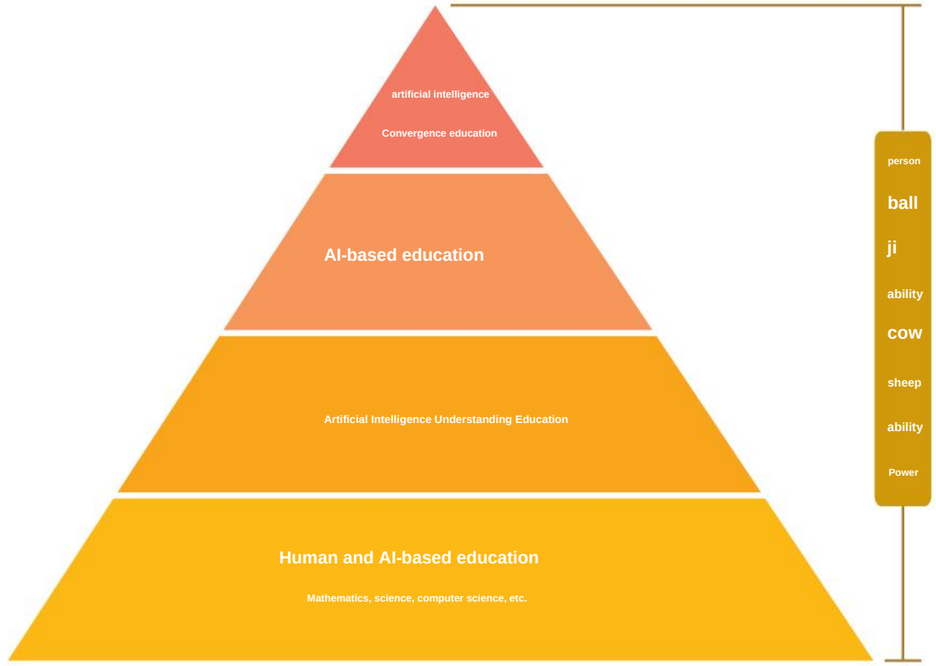
and the principles of algorithms.

Afterwards, practical capabilities will be developed through AI utilization education that applies AI technology to various real-world problems and utilizes it as a tool, and ultimately

It will expand into AI convergence education that creatively integrates AI with other fields to solve social problems and create new values.

In this model, AI education is not about acquiring fragmented functions, but rather about utilizing AI based on understanding and leading to creative integration.

It was emphasized that it was a holistic and progressive learning process.



[Figure] AI Education Model 1.0 (Korean Society for Artificial Intelligence Education, February 2020)

Machine Translated by Google

04

AI Education 2.0
Model Details

1 Why does AI basic social education require a new learning theory?

In the AI-based society, AI acts as a learning partner, a condition of existence, and a medium for restructuring social structures. Existing behaviorism, Cognitivism and constructivism emphasize "cognitive processing" or "experiential construction" based on human-centered thinking. However, AI performs learning, In an age of recommending, predicting, and sensory interaction with humans, learning now involves cooperation and sensitivity between beings, multi-layered connections and coordination, It must be redesigned in the context of co-creation of meaning and value.

Based on this new question about education, we propose two learning theories: meta-cooperationism and resonance.

ÿ How will we think, judge, and live with AI?

ÿ What balance will we create between technology and humanity, and what values will we choose?

ÿ What kind of beings should education cultivate in this complex world?

Meta-Collaborativism (Meta-Co Learning Theory)

This learning theory emphasizes continuous collaboration between humans and artificial intelligence (AI). Learners interact with AI based on metacognitive reflection. They collaborate to build knowledge and gradually complete tasks.

Core Principles

ÿ Human-AI collaboration: Learners perceive AI as a cognitive partner, not just a tool, and engage in meaningful interactions that deepen thinking and coordinate tasks. Participate in interactions.

ÿ Metacognitive awareness: Learners develop higher-order regulatory skills by monitoring and reflecting on not only their own thoughts but also the AI's responses.

ÿ Co-construction of meaning: Knowledge is not transmitted, but is jointly constructed through collaboration, verification, and repetition through dynamic interactions between learners and AI. is created with

ÿ Prompt-based scaffolding: Learners design, modify, and evaluate the prompts they input to AI, making the process itself meta-collaborative. Consider it as an activity and guide your thought process through it.

ÿ Distributed Cognition: AI can be used not only for repetitive and computation-based tasks, but also for higher-order cognitive activities such as idea generation, pattern analysis, and summarization. By engaging, learners collaborate with AI to distribute their thinking and co-create knowledge.

Traditional Cooperativism vs. Metacooperativism

item	Traditional cooperativism	metacooperationism
Equal human learners as learning subjects		Humans and AI are equal and complementary entities.
Process design roles are divided ÿ collaboration ÿ results are shared, problems are redefined ÿ AI and human joint exploration ÿ continuous feedback and adjustments ÿ new problems are discovered		

Machine Translated by Google

item	Traditional cooperativism	metacooperationism
Thinking level: Executive thinking focused primarily on problem solving Executive thinking + self-regulation and strategy design through metacognition and metathinking		
Relationship characteristics: short-term collaborative project-focused	Continuous and evolving relationship (symbiotic cooperation)	
Performance measurement task completion, cooperative attitude	Knowledge creation, adaptability, AI collaboration, and value judgment	

ÿ Based on the meta-cooperative learning theory, we construct an AI education 2.0 model, assuming that the AGI era has arrived in earnest.

Meta-cooperationism has deepened and developed into a learning theory of resonance.

Symbiotism

As a learning theory for the AGI era, it is an educational theory centered on the 'existential and cognitive co-evolution' between humans and artificial intelligence.

It's a paradigm.

Learners function as 'co-agents' based on self-reflection and social responsibility in a simultaneous evolving relationship with AI.

Create new knowledge, value, and meaning with us.

Core Principles

- ÿ Human-AI mutually evolving existence: Learners accept AI as an active partner, and both humans and AI interact and influence each other.

  We recognize ourselves as evolving beings. Learning is not a process of "information transfer," but rather a process of ontological mutual formation.

- ÿ Mutual accountability and ethical thinking: Humans are beings who take responsibility for the decision-making results of AI, recognize bias, and make ethical judgments, and are not simply

  They operate as 'co-designers' rather than users or supervisors.

- ÿ Ecological expansion of knowledge: Learning is ecological learning that takes place in real time in the network between humans, machines, and the environment, and knowledge is not static.

  It is organically created and expanded in a resonant flow, not as a stored object.

- ÿ Technology internalization and expansion of existence: AI is not an external tool, but is internalized as part of the learner's senses and cognition, and is integrated into the human way of being and learning.

  It changes the very method. Learning is not simply the use of technology, but an evolutionary extension of human capabilities.

- ÿ Formation of a resonant community of practice: Learners go beyond being a single entity collaborating with AI to become a community of practice where AI, humans, and society grow together.

  Become a member and actively participate in joint problem solving, value creation, and designing the future society.

<Table> Comparison of learning theories by era

<Table> Comparison of learning theories by era

Concepts central to contemporary learning theory	Knowledge Center	Teacher's role	Learner Role	Philosophical background
Industrial society behaviorism stimulus-response extrinsic knowledge acquisition		communicator	passive recipient	positivism, Mechanistic view of humanity
Information society, cognitivism, information processing, internalized structure, structural explainer, active information processor, rationalism, cognitive science				
network society	Constructivist meaning construction	Contextual composition facilitator/guide	meaning creator	Constructivist epistemology, Social and Cultural Philosophy
AI	Meta-collaboration caution	Metacognition-based Human-AI interaction Action and cooperation	Co-creation in relationships and reflective composition	Promoting meta-collaboration Self and designer	Interacting with AI The reflective meaning through co-creator	Dialogic epistemology Distributed Cognition metacognitive theory Postcognitivism
Basic Social Studies	resonance	Co-evolution of Humans and AI	The unity of existence and learning Synthetic co-evolution	Ecological Systems Designer/Ethical Facilitator	AI and humans interact Learning dependently Evolving together existence	Posthumanism, Relational ontology

2 What is digitalization?

Digitacy is a compound word of digital and literacy, and simply refers to the ability to handle digital devices well or to use ICT.

Beyond that, it means the ability to apply computer science-based technologies and computational thinking to real-life problem-solving processes.

Digital literacy ÿ digitality
Digitacy = Digital + Literacy

Existing digital literacy focuses on the information generated by digital devices, emphasizing the collection, communication, evaluation, and ethical aspects of information.

Therefore, the original purpose of computer science is not to be considered an academic discipline, but rather a concept of general knowledge that deals with language and social fields that emphasize communication.

Considered.

However, digital skills are not just about the ability to use digital devices; they are about the ability to apply computer science to real life.

This can be understood in a similar context to how counting has developed beyond simple calculation skills into numeracy.

Just as arithmetic has evolved to include mathematical thinking and practical application skills in response to the demands of the times, digital literacy also encompasses computing.

Machine Translated by Google

Developed into a more comprehensive and in-depth concept and established under its own name to adapt to the rapidly changing technological environment of today and the accident (CT) era.

Because it has to be done.

This is not limited to technical proficiency, but rather involves critically adapting to the current digital environment and preparing for the future through a balance of knowledge, skills and attitudes.

It can be said to be a talent for taking the lead actively.

The need for digitalization

ÿ Requirement for adaptation and initiative in a rapidly changing technological environment

ÿ It is difficult to cope with the complex and advanced digital environment with only existing technical literacy. Therefore, it is necessary to proactively adapt to technological changes.

It is essential to develop systematic capabilities to respond and adapt effectively to new environments.

ÿ The ability to accept and actively lead change, rather than simply adapting to the digital environment, is required.

ÿ Digital technology is not static knowledge, but a constantly changing field. Anyone who does not continuously learn will be digitally excluded.

It can be hierarchical.

ÿ Digital literacy is not a skill reserved for a specific class, but rather a universal quality centered on lifelong learning that all citizens must continuously develop.

ÿ Overcoming the limitations of existing digital literacy and media literacy capabilities.

ÿ Many citizens in modern society lack understanding of the fundamental principles, structure, and concepts of digital technology, and are only focused on ethical discourse.

Digital education reveals limitations in expanding technological capabilities.

ÿ While only the knowledge of ethical issues related to digital technology is emphasized, the basic knowledge and skills required to understand and utilize technology are neglected.

To address these shortcomings, digital technology must be comprehensively taught, from the basics to advanced knowledge.

Do it.

ÿ We need to move beyond a digital ethics and critical approach and strike a balance in creating opportunities using SW and AI technologies.

ÿ The approach to digital literacy in media literacy can hinder the development of technology itself and create a utopian world through technology.

It can lead us to ignore the potential. Digitasi aims to correct this bias and balance the positive potential of technology with its ethical responsibilities.

It should provide core competencies that can be handled. Digitality should be considered not only for the ethical aspects of technology, but also for the opportunities that can be created through the technology.

Focus on exploring and exploiting possibilities.

ÿ Establishing literacy and math skills as core competencies in a digital society

ÿ In modern society, literacy, numeracy, and digitacy are the three most important literacies.

Literacy encompasses traditional reading and writing skills, while numeracy emphasizes basic mathematical thinking and problem-solving skills.

In addition, digital city focuses on developing the complex and in-depth capabilities required in a digital environment.

ÿ Beyond simple tool use, we must comprehensively cultivate capabilities such as information processing, programming, and AI utilization.

ÿ Requirement of real-life problem-solving skills

ÿ In a similar context to how mathematical ability has expanded into the ability to apply mathematics in real life, digital technology has also become a practical tool that contributes to solving real-world problems.

It should work.

ÿ Digital City is not just about simple technical knowledge, but also about being able to use technology in real life to solve problems and seize technological opportunities.

It must develop into ability.

Mathematical thinking and numeracy

Mathematical thinking is the ability to understand and reason about the essence of mathematics, and is 'thinking within mathematics'.

Mathematical literacy is the ability to read, interpret, and apply mathematics in real life, and is 'thinking about life through mathematics.'

ÿ Mathematical thinking is the ability to think deeply about mathematical problems and solve them logically.

ÿ Numeracy is the ability to apply mathematics to life problems and make rational decisions.

division	mathematical thinking skills (Mathematical Thinking)	Numeracy / Mathematical Literacy
concept	The ability to think logically and abstractly and solve problems using mathematical concepts and structures.	The ability to understand, judge, and make decisions by utilizing mathematical knowledge and skills in real-life contexts
focus	The thinking of mathematics itself (problem solving, argumentation, abstraction, generalization, etc.)	Applying mathematics to real-world problems (Information interpretation, quantitative judgment, literacy-based utilization, etc.)
Objective: Deep understanding of mathematical concepts, logical thinking, real-life problem solving, and rational decision making.		
Emphasis on higher-order thinking skills in background mathematics education		Practical literacy emphasized in international assessments such as the OECD PISA
Focused on abstract and theoretical problem solving		Context-based and practical application-focused
example	Infer, generalize, and prove the properties of functions "What mathematical principles are at work in this problem?", "Does it hold in all cases?"	Interpreting statistical graphs in newspapers, calculating product discounts, budgeting, etc. "Is this graph reliable?", "Is this price actually profitable?"

Machine Translated by Google

Informational thinking and digitality

Informational thinking is the ability to logically analyze and process information and solve problems based on the principles of computer science.

DIGITACY goes beyond computational thinking and extends it to the ability to apply and practice it to real-life and social problems.

It is a concept that has been established. In other words, if information thinking focuses on 'how to structure and process a problem,' digital thinking focuses on 'that ability.'

It focuses on 'how to live and use it in life and society.'

ÿ Informational thinking is a technology-centered thinking ability that designs how to organize and process information by thinking like a computer.

ÿ Digitality is the ability to live wisely in a digital society, not only understanding technology but also taking responsibility for real-life problems.

The ability to apply and practice

division	Informatics Thinking	Digitacy
concept	Informatics-based problem-solving thinking: A way of thinking that analyzes, structures, processes, and utilizes information	Critical, creative and responsible information and communication in a digital environment Practical skills to handle technology
focus	The ability to recognize and solve problems based on information (including information processing, algorithms, automation, etc.)	Basic literacy skills for citizens to survive in the digital society, That is, it includes digital thinking, communication, ethics, and participation capabilities.
	Organize and utilize objective information logically to solve problems	Understanding the principles of technology in the digital age and being critical and ethical Living as a user/producer
	Characteristic instrumental and technical thinking (information analysis, processing, automation, etc.)	Focus on practical and social competencies (information judgment, creative expression, collaboration, digital citizenship)
	Application areas: Information processing, algorithm design, information system utilization, etc.	Digital communication, platform utilization, digital ethics, online Identity, etc.
	Origins: Information science, computer science, and computational thinking	The evolved concept of digital literacy (literacy (Poetry + Digital Technology Understanding + Practical Sensitivity + Social Participation)

Domains and sub-elements of digital city

ÿ Criteria for selecting digital areas

ÿ Real-life applicability: Can you use digital tools or data to solve problems in your daily life?

ÿ Computer Science Foundation: Are computer science concepts such as algorithms, hardware, and networks reflected?

ÿ Digital Adaptability: Can you analyze, select, and create information in a rapidly changing digital society?

ÿ Problem-solving structure: Is the logical structure of problem recognition ÿ analysis ÿ solution creation revealed?

ÿ Civic Responsibility and Ethics: Are the rights, responsibilities, and ethics of digital participants reflected?

ÿ Principles for configuring digital sub-elements

ÿ Concept-Function-Attitude Integration: Each sub-element includes knowledge understanding, practical activities, and value judgments.

ÿ Hierarchical structure by school level: The depth of concepts and level of practice are divided into stages according to the developmental level of elementary, middle, and high school.

ÿ Securing connectivity between literacies: Consider the interactive structure of literacies such as information, data, and AI (e.g., search ÿ evaluation ÿ utilization ÿ ethics).

ÿ Maintaining technology neutrality: It is structured around principles and is not dependent on specific devices or software.

ÿ Based on real-life situations: All sub-elements are composed of content that can be applied to real-life problem situations.

<Table> Seven Digital Literacy Areas and Their Objectives

Large area	Composition goals
digital	Understand and explain how digital systems and technologies work.
ICT literacy	Ability to manipulate and utilize various digital devices and communication technologies for specific purposes.
Information literacy	Information can be searched, evaluated, processed, and stored for a specific purpose and used to solve real-life problems.
Computing Literacy	Problems can be analyzed procedurally and solved using algorithms and programming.
data literacy	You can collect, analyze, and visualize data and make decisions based on it.
Digital Citizen Literacy: Understand and practice ethics, security, rights, and responsibilities in a digital society.	

<Table> Digital Achievement Standards by School Level

Large area	sub-elements	Elementary	Secondary	High school
Digital understanding	Bits and Bytes Understanding the concept	Compare file sizes or save space In situations where the liver is insufficient, beets and Solve the problem by using the byte concept I can explain.	When saving or transferring files, the data Understand the size of the data and be efficient I can explain how to use it.	For data transmission, compression, storage, etc. Utilizing bits and bytes To analyze efficiency and performance Can be.
	Binary and logic Understanding operations	Activities that express information using 0 and 1 The operation of digital devices through You can experience Lee.	Understand the meaning of logical operations, To solve the conditional judgment problem Can be utilized.	Using binary and logical operations Here, in digital circuits or condition processing Analyze and predict the behavior of the stem Can be.
	Computer components Understanding cows	The main components that make up a computer Can explain names and roles.	CPU, memory, storage, etc. Understand the role, performance and purpose Select the appropriate device according to Can be.	The main components and interactions of a computer Understand the function and purpose A system that takes performance and cost into account Can be designed.
	Operating system functions and understanding the role	Run the app through the operating system You can manage folders.	Understand the basic functions of the operating system High, file management or multitasking issues It can be applied to the solution.	Operating system resource management, multitasking King, user interface, etc. and apply it to solving real problems Can be.

Machine Translated by Google

Large area	sub-elements	Elementary	Secondary	High school
ICT Literacy	digital devices Manipulation and utilization	Using a smart device or computer Activities that involve finding or expressing information can be used to solve problems.	Characteristics and functions of digital devices Teach and use appropriately according to the purpose It can be used.	Various digital devices and functions Compare and analyze to find the right solution for the problem situation Select and utilize the combined device Can be.
	Cloud Tool utilization	Keep your files safe and secure with your friends Cloud storage to share with You can use cows.	Share or collaborate on files Cloud-based collaboration tools for can be used.	Depending on the purpose of collaboration, backup, and sharing Choose from a variety of cloud tools Increase productivity and collaboration efficiency Can be.
	Network Basics and connection structure understanding	How the Internet is connected Knowing the simple flow, at home or at school Use your device with Wi-Fi Can be connected.	The difference between the Internet and Wi-Fi and simple network connection I can explain the method.	Networking such as IP addresses, DNS, routing, etc. Understand the basic concepts of the My network problem is structurally It can be diagnosed and resolved.
	IoT and communication technology Understanding the concept	Smart phones used at home or school Look at your smart devices (e.g. smart lights, wireless speakers, etc.) and see if they are compatible with your device. How they are connected and work I can talk about it.	The operating principles of IoT devices in daily life Describe the components and how to use them Examples can be analyzed.	IoT system structure and sensor network Set up data flow between work servers and provide real-life application examples. It can be done.
information Literacy	Information retrieval strategies and use of search tools	You can find information on the Internet by specifying search keywords.	Appropriate keywords, search terms, and filters Use it to effectively find the information you want It can be searched and refined by law.	Advanced search queries for complex topics and information using various tools To provide results that are suitable for the quality and purpose of Can be released.
	Reliability of information and bias assessment	Information from various sources Why is one more trustworthy? I can explain it to you.	Compare sources of information and their reliability can be evaluated.	Considering the information producer and context Reliability, accuracy, and bias of content Evaluate and critically utilize gender You can do it.
	Summary of information and Reconstruction	Read various materials and understand the key points Organize or reorganize into presentation materials You can do it.	Structure and present the collected information Reconstructed into data or report format You can do it.	Summarize vast amounts of information for your purposes Reconstruct and reuse visual materials or programs In various formats such as presentations Can be delivered.
	Information storage and Recycling	Save useful information to a folder And, I will find it again when needed Can be used.	Categorize information by topic, and Use it appropriately considering copyright Can be.	Considering copyright and information ethics Structure information and keep it in the long term Save in a usable way You can do it.

Large area	sub-elements	Elementary	Secondary	High school
Computing Literacy	Algorithm Design and Expression	Step by step solving everyday problem situations We have organized the procedures so that we can conclude I can explain.	Contains various conditions and repetitions Analyze the problem procedurally, Block efficient solution algorithm It can be implemented through coding.	Analyze the problem in various ways and considering efficiency and scalability Design and implement an algorithm Can be.
	Block coding and Text Coding Basics	Simple programming through block coding You can make RAM.	Simple using block coding tools One program can be implemented.	Text-based programming language Utilizing input-processing-output structure Write and debug programs You can do it.
	Condition/Repeat Structure utilization	or behave differently depending on the conditions In situations where repetitive tasks are required Here is a solution using block coding You can implement it inside.	Using conditional statements and loops You can design a rational program. there is.	Nested conditions and repeating structures Use it to create complex programs Can be.
	Debugging and Problem-solving strategies	Desired result while block coding When it doesn't come out, find the problem You can try to fix it.	Analyze the cause of the program error So, we can solve it by setting up a correction strategy. there is.	Distinguish between logical and execution errors and debugging tools and strategies. Solve problems systematically using You can do it.
data Literacy	data Collection and Refining	On topics of interest in life Gather information and avoid unnecessary You can filter out data.	Collect data according to purpose High, clean up errors and duplicate data To create reliable data Can be.	Cleanse data to suit your purpose It can be classified by type.
	data Analysis and Visualization	Using tables and graphs, you can share your friends' Compare and explain opinions or situations You can do it.	Use tables and graphs to present data Can be compared.	Utilizing statistical techniques and visualization tools Analyze the data and make it meaningful can derive insights.
	data-driven Decision making	Which of the various materials is more appropriate? Select and judge based on the grounds You can do it.	Based on data, appropriate to the situation can make decisions.	Depending on the situation, various data are displayed Analyze and provide evidence-based recommendations You can make enemy decisions.
	Data security and Privacy	Among the articles or photos posted on the Internet Determining what is personal information You can be careful.	The meaning of personal information and how to protect it Understand and use data safely You can put the rules into practice.	In the process of data collection and utilization Recognize emerging security and ethical issues And according to the personal information protection standards You can respond.

Machine Translated by Google

Large area	sub-elements	Elementary	Secondary	High school
digital citizen Literacy	Digital etiquette and sense of community	Be considerate of your friends, even in online spaces And you can communicate politely.	Communication and collaboration in online communities Be polite in situations and in public I can fulfill my responsibilities.	Empathy, responsibility, and trust in the digital environment Based on the lightning, the community members Can perform the role.
	Personal information Protection and Security	Know the importance of personal information and protect it can practice this method.	Password management, public settings, etc. through yourself in the digital space Can protect.	Personal information according to the development of digital technology Understand and prevent the types of security breaches and can establish response strategies.
	cyberbullying Prevention and Response	Bad comments or teasing posts hurt Knowing that it could be, What to do in this situation It can cause.	Types of cyberbullying and how to respond to it Know and respond appropriately when damage occurs The plan can be implemented.	The structure and severity of cyber violence Recognize, prevent and resolve damage Proposing social response measures for You can do it.
	Digital society and Understanding Career Change	With the advent of digital technology, new A job that has changed or a job that has been changed Look at the humidity and find out why You can talk to the people.	Jobs Changed by Digital Technology You can investigate the world.	Jobs brought about by digital transformation Structural changes in the world and the needs for them Analyze your capabilities and reflect on your career path It can be done.

<Table> General Citizen Digital Achievement Standards

Large area	sub-elements	citizenry
digital	Bits and Bytes Understanding the concept	Understand and efficiently understand the concepts of bits and bytes in the context of storing, transmitting, and compressing digital data. You can make a choice.
	Binary and Understanding logical operations	How binary and logical operations are applied to the operation of smart devices and automated systems. I can explain why.
	computer Understanding the Components	Understand the roles and performance of CPU, memory, storage, etc. to select devices or diagnose problems. You can do it.
	of the operating system Understanding functions and roles	Understand the basic functions of the operating system (file and app management, etc.) and respond appropriately when errors occur. Can be.
ICT Literacy	digital devices Manipulation and utilization	You can choose a digital device that suits the situation and use it efficiently according to the purpose.
	Leverage cloud tools	You can choose and utilize cloud-based tools for document creation, collaboration, and backup.
	Understanding network fundamentals and connection structures	Check your Wi-Fi, IP address, internet connection status, and fix simple network problems. You can do it.
	Understanding IoT and communication technology concepts	Understand the operating structure of IoT devices such as smart homes and wearables, and determine their safety and efficiency. You can do it.

Large area	sub-elements	citizenry
information Literacy	Information retrieval strategies and use of search tools	Efficiently find the information you want using keywords and search strategies that fit your purpose. You can do it.
	Assessing the reliability and bias of information	Critically judge the reliability and bias of information by analyzing its source, purpose, and method of expression. You can do it.
	Summarizing and reorganizing information	It can summarize complex information or reorganize it to suit the purpose and make it easier to convey.
	Information storage and recycling	Information can be stored and reused appropriately, taking into account the copyright and usage conditions of the information.
Computing Literacy	Algorithm Design and Expression	Repetitive tasks or problem situations can be solved by procedurally analyzing them and automating or structuring them.
	Basics of block coding and text coding	Create simple problem-solving programs using simple automation tools or coding education platforms. Can be.
	Condition/Repeat Structure utilization	You can recognize conditions and repetitive flows and apply them to automation in your daily life or work processes.
	Debugging and Problem-solving strategies	When an error occurs while using a digital tool, the problem is diagnosed and corrected according to a logical procedure. Can be decided.
data Literacy	Data collection and cleaning	Collect the data you need in your daily life or work, remove errors and duplication, and provide reliable data. You can organize the data.
	Data Analysis and Visualization	You can interpret data and derive meaningful information using tables, graphs, statistics, etc.
	Data-driven decision making	You can compare and analyze data to make rational decisions in real-life or work situations.
	Data Security and Privacy: Safely protect personal and sensitive information and practice data ethics.	
digital citizen Literacy	Digital Etiquette and Community Sense	Communicate with consideration and responsibility online and contribute to the formation of a healthy digital community. You can do it.
	Personal information Protection and Security	You can protect your own and others' personal information and respond to risks such as phishing and hacking.
	cyberbullying Prevention and Response	Recognize cyberbullying and hate speech and take proactive action to prevent and address it. You can do it.
	Digital society and Understanding Career Change	Understand and proactively respond to the changing work environment and job structure driven by digital technology.

Machine Translated by Google

3 What is AI thinking?

Defining AI Thinking

"The ability to think collaboratively with AI to solve problems."

AI thinking is based on existing computational thinking, but fundamentally changes the scope, target, and method of thinking.

It has been expanded to .

In other words, if computational thinking is "a way of thinking that transforms human-centered thinking ability into computer-based problem solving," AI thinking is its extension.

In line with this, we are going beyond simply 'thinking using a computer' and are changing and expanding the structure of thinking itself by collaborating with an intelligent partner called artificial intelligence.

It's a way of doing business.

AI thinking ability is based on the ability to understand and reflect on the operating principles and judgment structure of artificial intelligence, data-based reasoning, and limitations, and solve problems.

It includes higher-order thinking skills that enable us to redefine, collaborate with AI to explore solution strategies, and make responsible judgments about outcomes.

This is not simply a technical understanding, but a relational thinking ability in which humans and AI think together. The components of AI thinking include:

There are four types: artificial intelligence, automation, intelligence, and autonomy.

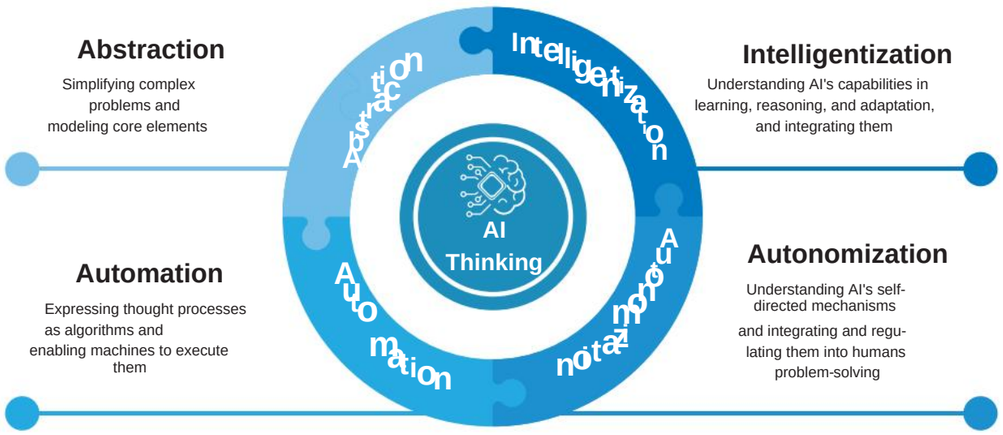
ÿ Abstraction: The ability to simplify a problem and express its essence.

ÿ Automation: The ability to transform thought processes into actionable ones through algorithms and code.

ÿ Intelligence: Understand AI's learning, reasoning, cognition, prediction, and adaptation capabilities, and integrate them into problem solving to enable humans and AI to work together.

The process of making judgments and higher-level decisions

ÿ Autonomization: The ability to design and coordinate AI and humans to jointly perform autonomous judgment and execution.



"Structure the problem into its core (abstract), design it to be executable (automate), and integrate AI's judgment and learning (intelligence),

"Higher-order thinking ability that enables humans and AI to think and act autonomously together"

Core Elements of AI Thinking

1. ABSTRACT: The process of simplifying complex reality and revealing the essence of the problem.

ÿ The ability to remove unnecessary elements from real-world problems and express only the core structure.

ÿ Classify and model diverse data to enable AI to recognize and process it.

ÿ Thinking that expresses human thought systems in language, numbers, diagrams, etc. and creates a format that can be interpreted by machines

2. Automation: The process of designing a thought process into an algorithm and having it executed by a machine.

ÿ Ability to logically and implement abstract problem structures into algorithms or procedures

ÿ Automate problem-solving strategies through coding, rule design, and automated repetitive flows.

ÿ Securing not only feasibility through programming, but also proceduralization and reproducibility of thinking

3. Intelligence: Understand the intelligent thinking structure of artificial intelligence, such as learning, reasoning, recognition, prediction, and adaptation capabilities, and apply it to problem solving.

A process in which humans and AI collaborate to judge, adapt, and perform higher-level thinking and decision-making by integrating into the solution process.

ÿ Understand and apply the intelligent operating principles of AI, including its judgment process, learning structure, and reasoning method.

ÿ Leverage AI's cognitive capabilities to predict, classify, and judge large-scale data that is difficult for humans to process.

ÿ Solving higher-order problems by combining human sensory and intuition-based thinking with AI's analysis and prediction-based thinking.

4. Autonomization: The process of designing AI and humans to jointly make decisions and execute autonomously according to the context.

ÿ Ability to design collaborative structures between AI and humans and solve problems through continuous interaction

ÿ Do not blindly accept the results of AI execution, but adjust them by considering human goals, values, and responsibilities.

ÿ Supports AI systems to make decisions, adapt, and execute on their own according to the situation, and humans supervise or cooperate with them.

[Prerequisites for Autonomy]

Humans do not delegate full autonomy to AI, but rather function mostly as designers, coordinators, and interpreters. Therefore, in AI's thinking ability,

The "autonomy" we speak of is "cooperative autonomy (co-agency)" and contributes to the problem-solving process that is valuable to humans.

Machine Translated by Google

AI Intelligence at the Current Level

area	explanation	AI level	example
Perception	Various types of images, voices, texts, etc. Detect and interpret an input	Human level or higher (specific (Transcendent in work)	Image classification (YOLOv8), sound Gender Awareness (Whisper), Translation
Memory	Store large amounts of information and quickly Search for lege	Very good (accuracy and speed are (transcends the liver)	Context of large-scale language models Processing, vector search
Pattern Recognition Recognition)	Rules and patterns in massive data Extract the turn	Excellence (strong with structured data)	Recommendation systems, fraud detection, and Series prediction
Language Understanding Understanding)	Sentence interpretation, context understanding, questions response	Practical at the GPT-4o level Conversation is possible, but still Lots of sun	ChatGPT, Claude, Gemini
Reasoning	Premise ÿ Argument leading to conclusion logical thinking	Good at simple logic, but complex The reasoning is inadequate	Solve some chess and math problems
Situational judgment and autonomous decision-making (Autonomous Decision-Making)	Goal setting based on contextual understanding and execution coordination	Limited, human feedback still available Hi essential	Self-driving cars, robotic arm control, self Same response
Emotions and ethical judgment (Affective & Ethical Intelligence)	human emotions or value judgments consideration	Very inadequate, within the learned framework West operation	Sentiment analysis, algorithmic bias My occurrence
Creativity	New ideas, content creation castle	Recombination creativity based on existing learning Sung is excellent, true creativity Silver limit	Image creation, composition, and scenario Drafting

AI Autonomy at the Current Level

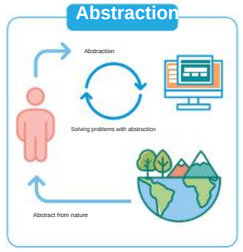
Intelligent functions	explanation	Level of autonomy
perception	Recognize images or convert speech to text	height (high accuracy)
Expression/Understanding	Process natural language questions and perform summarization and classification	middle (limited understanding of context)
inference	Rule-based judgment, statistical prediction, and sentence generation in LLM	Limited (not really 'understanding')
Learning/Adaptation	Some systems are fine-tuned to reflect user feedback.	Limited (fine-tuning level)
Behavioral/Communication	Recommendation Systems, Conversational AI, and Autonomous Driving	Conditional autonomy (Scenario-based execution)
Entrusting important judgments and decisions in autonomous decision-making/decision-making problem-solving to AI.		

* Currently, the autonomy of AI can be said to be 'a level where it can perform some judgments within the boundaries set by humans under a set goal.'

Machine Translated by Google

<Table> Abstraction, Automation, Intelligence, and Autonomy

Abstraction



Abstraction is the process by which humans observe nature and the world and identify meaningful patterns and structures within them.

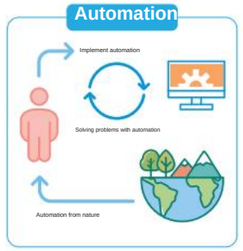
It is a thought process that draws out the core elements. People simplify the complex phenomena of reality and

Reconstruct the core structure of the problem. This allows AI to recognize and process real-world problems.

It is a stage of rock expression, and the beginning of AI thinking is always from this 'sensation and expressiveness that penetrates the essence'.

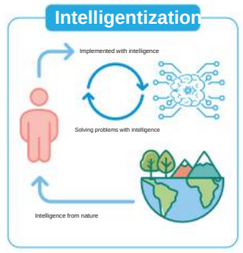
Departure.

Automation



Automation is the thought process that transforms abstract problems into mechanically executable steps. Between humans and machines, the implementation of automation and problem solving through automation are circulating, demonstrating that human-derived structures are designed into computer algorithms and systems to be executed in a repeatable manner. In other words, automation is the proceduralization of thinking (procedure design), and it is the link that ensures that the way humans think is consistently implemented through technology.

Intelligentization



Intelligence goes beyond simple repetition of automation, and creates new ways of interacting with AI.

It is a high-level thinking process that solves problems and enables AI to learn and evolve.

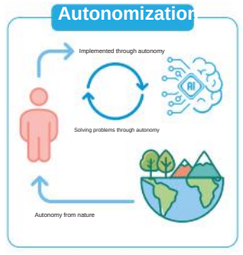
The continuous cycle between people and AI is emphasized in the problem-solving process, which means that humans can solve problems through AI.

At the same time, AI also expresses a structure that learns from nature and humans and becomes intelligent.

At this stage, humans are not just using AI, but thinking and co-creating with it.

Becomes the subject of .

Autonomization



Autonomy is the level at which human-designed algorithms and data-driven models simply carry out commands.

Beyond that, it is designed and operated to make optimal choices on its own according to changes in the external environment.

It means that it is learning, reasoning, perception, and prediction.

It includes high-level functions such as (prediction). Humans accept or control the autonomy of AI.

Collaborating, AI assists or leads humans in solving complex problems, and increasingly becomes an intelligent partner.

It's angry.

References

Expansion (augmentation) of capabilities through AI thinking

AI thinking is not simply the ability to "think well with machines," but rather the ability for humans to reflect on their own thought processes (meta-thinking).

A level of cognitive leeway and insight that can interpret and adjust AI's judgment must be presupposed.

1. Metacognition

The ability to understand why the results presented by AI came out that way and to reflect on them on your own, rather than blindly accepting them.

2. Data-driven predictive awareness

Thinking that forecasts and predicts problems based on AI output patterns and adjusts strategies

3. Critical Receptivity

The ability to recognize and critically accept the limitations and possibilities of AI, rather than uncritically accepting the results of AI.

4. Human-AI Collaboration & Communication

The ability to communicate effectively with AI or coordinate collaboration between people and AI.

5. Creative Transferability

The ability to creatively transfer and apply a single AI concept or principle to various contexts and problems.

6. Algorithmic Problem Solving

Understand AI's thinking process and be able to think and process algorithmically when solving human problems.

7. Ethical Reflexivity

The ability to sensitively reflect on the ethical implications of the results generated by AI or the data used, and to make judgments and act accordingly.

8. Sensory Imagination

The ability to understand how AI processes sensory data (e.g., images, sounds, emotions, etc.) and connect this to human sensory experiences to reason about it.

9. Systems Thinking

The ability to understand the impact of AI technology on individuals, society, and the ecosystem from a macroscopic perspective and to think within the overall system.

Machine Translated by Google

References

FAQs About AI Thinking

1. Human Autonomy: When Is It Possible?

condition	explanation
Cognitive awareness is when I can consciously reflect on what I am doing and why I act that way.	
Subjectivity of judgment: When you can judge and choose for yourself without being swayed by external influences.	
Value-based thinking: When the standards for behavior are based on social values or ethics.	
Reflective thinking: When you can look back on your own judgments and results and improve them.	

Example: Autonomy is in effect when a student does not blindly accept AI-recommended books, but rather adjusts their own selection based on their reading preferences and learning goals.

2. Autonomy of Machines (AI): How Does It Happen?

element	explanation
Environmental awareness	Status detection using sensors, data input, etc.
Targeting through external setting or reinforcement learning	
Applying decision-making models optimized for judgment algorithms (e.g., MDP, reinforcement learning, etc.)	
Learning and tuning: Ability to adjust behavior through trial and error and feedback (e.g. AutoML, AutoGPT)	

Example: A self-driving car automatically adjusting the speed limit or taking a detour to the destination by judging the traffic situation is the result of autonomous judgment and execution.

3. Educational Implications: What does learner autonomy mean?

Beyond simply 'using' AI, humans have the ability to understand, interpret, and adjust AI's judgments, that is, to create value beyond the functions of AI.

It is about becoming a thinking subject who can make ethical judgments and choices.

- The challenges of education

ÿ The ability for students to make their own decisions without relying on AI

ÿ The ability to recognize the limitations and risks of AI and utilize them appropriately

ÿ Internalizing the collaborative perspective that "AI helps me solve my problems."

References

Comparison of computer literacy, computing skills, and computational thinking

1. Computing Literacy

ÿ Understand the basic concepts of computers and information technology and acquire basic skills that can be applied in daily life and learning.

ÿ Basic understanding of computer architecture, operating systems, the Internet, file storage methods, and information ethics.

ÿ Includes practical skills such as Word, Excel, coding experience, and basic AI experience.

ÿ Focus on 'Do you know it?' and 'Can you use it?'

2. Computing Competency

ÿ Practical ability to solve real-world problems by utilizing computer-related tools, languages, and systems

ÿ Includes coding, algorithm design, data structures, data analysis, and network utilization.

ÿ Emphasize practical feasibility and applicability

ÿ Can you do it? That is, can you give a problem and have it solved?

3. Computational Thinking

ÿ Cognitive ability to understand problems using computational thinking, structure them, and design solution strategies

ÿ Thinking processes such as problem decomposition, pattern recognition, abstraction, algorithms, and automation

ÿ No programming or coding required

ÿ Meaning a shift from 'human-centered thinking ability' to 'computer-based problem solving'

division	Computational Thinking (CT)	AI Thinking (AIT)
Base transition	human thought ÿ Computer-based problem solving	computer-based thinking ÿ Expanding thinking to collaborate with AI
Central concepts: problem decomposition, algorithms, and abstraction		Judgment, learning, prediction, adaptation, cooperation
Instrumental thinking: Computers are execution tools		AI emerges as a judging, recommending, and collaborative entity.
The nature of thinking is procedural and centered		Focused on contextual, adaptive, and collaborative thinking
human role	Design the problem Leave the execution to the machine	Critical of AI's judgment Accept and coordinate together

Machine Translated by Google

4. Learners' cognitive level improved by AI thinking skills

AI thinking ability is different from the existing cognitive taxonomy of Bloom, ranging from data-driven sensory cognition to predictive thinking and ethics. It involves an extended thought process that leads to critical reflection. In particular, AI automates the process of detecting, classifying, and predicting data. Structuring the information that forms the basis, examining the legitimacy of judgment, and giving it human meaning are still the learner's responsibility.

Accordingly, AI Education 2.0 organizes the learner's cognitive development process into seven stages.

ÿ Perceive ÿ Explore ÿ Comprehend ÿ Reason ÿ Apply ÿ Predict

ÿ The flow of reflection (Metacognize) is how AI processes information and how humans interact with it and expand their thinking. It shows.

This cognitive process is not a one-way staircase, but rather a spiral that is constantly revisited and expanded. Learners are AI. Rather than passively reacting to the information provided by AI, AI understands, manipulates, reflects, and actively reconstructs its thoughts. Grow as a subject.

AI thinking is key to fostering the capacity of citizens to reconstruct knowledge and create value with AI, rather than being knowledge consumers in the information society. am.

7 Cognitive Levels Required in AI Education 2.0

The 7 cognitive levels required in AI Education 2.0 are the learning difficulty levels of the learner's cognitive thinking development process that is enhanced through AI thinking ability. And it can be divided according to the learner's growth.

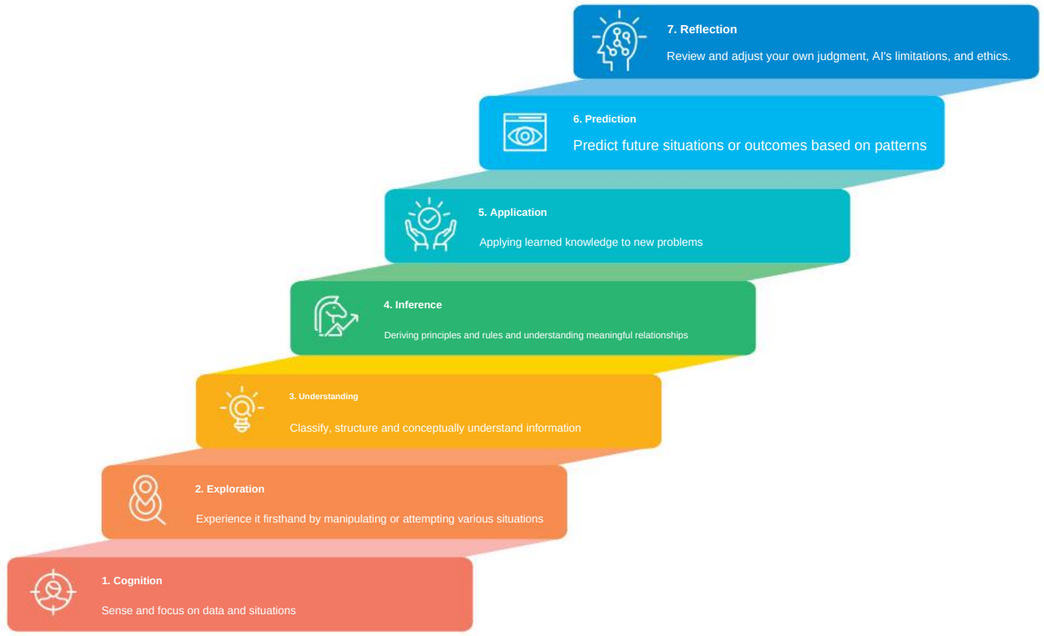
First, the lower levels (1-3) of Perceive, Explore, and Comprehend are the process of AI receiving data and responding to it. This is the stage where learners perceive information sensorily, experience it in various ways, and structure it to conceptually understand it. This stage It involves the basic thinking processes by which learners perceive the external world and internalize information.

Next, the intermediate stages (stages 4 and 5), Reason and Apply, correspond to the process in which AI processes information and makes decisions. Learners will demonstrate the ability to perform meaningful logical reasoning based on given information and rules and apply this to solving real-world problems. This step serves as a pivotal point for expanding thinking and plays a key role in performing thinking with AI.

Finally, the upper stages (6-7) of Predict and Metacognize are where AI generates output values and uses them to improve itself. It is connected to the process of correction. Learners simulate future situations based on data and experience, predict outcomes, and make decisions based on AI. It develops the ability to critically reflect on one's own thoughts. This goes beyond the simple use of knowledge and develops higher-order cognition and ethical judgment. It requires a sophisticated level of thinking that encompasses everything up to and including .

<Table> 7 Cognitive Levels for AI Education 2.0

step	Key Concepts	AI response	educational response
ÿ Cognition (Perceive)	Sense and perceive data or situations Focus	Sensor input, data detection	Observation, sensory capture
ÿ Exploration (Explore)	By manipulating or trying out various situations Direct experience	Diversify input, initialize simulation	experiential learning
ÿ Understanding (Comprehend)	Classify, structure, and conceptually understand information	Feature extraction, vectorization	Conceptual diagramming and classification skills
ÿ Inference (Reason)	Rule-based judgment and conditional inference to identify semantic relationships by deriving principles or rules		Logical thinking, understanding of principles
ÿ Application (Apply)	Applying learned knowledge to new problems	Apply the model and perform tests	Problem solving, application skills
ÿ Prediction (Predict)	Deep learning predictions and outputs based on patterns to predict future situations or outcomes.		Simulation, predictive thinking
ÿ Reflection (Metacognize)	Review your own judgment, AI's limitations, and ethics. High adjustment	Feedback loops, self-optimizing reflection, ethical thinking, critical thinking	



[Figure] AI Education 2.0 Cognitive Level Hierarchy

Machine Translated by Google

Achievement Standards Based on Learner Cognitive Level in AI Education 2.0

The achievement standards for AI Education 2.0 according to the learner's cognitive level reflect the core competencies common to elementary, middle, and high school, and can induce gradual development. It was designed around behavioral indicators.

<Table> Achievement Standards by Learner Cognitive Level

step	Achievement standards
recognition (Perceive)	Learners can observe various types of data and understand how AI perceives them.
quest (Explore)	Learners can directly manipulate AI-related tools and simulations to experience data input methods and processing flows.
understanding (Comprehend)	Learners can categorize or visualize data to explain the features and structures that AI uses for classification.
inference (Reason)	Learners can analyze the basis for AI's judgment and logically explain what results will be derived under given conditions.
apply (Apply)	Learners can apply the AI principles and tools they have learned to real-world problem situations and evaluate the results.
prediction (Predict)	Learners can simulate or explain future outcomes that AI might predict based on data.
Reflection (Metacognize)	Learners can compare AI's decision-making process with their own judgments and critically reflect on its limitations and ethical issues.

<Table> Achievement standards for elementary, middle, and high school students by cognitive level

step	Achievement standards		
	Elementary	Secondary	High school
recognition	AI detects surrounding objects or images Find elements that can be used (color, shape, etc.) I can tell.	How AI processes different types of data I can explain what I am sensing.	The process of AI sensors detecting data is real. This can be explained with an example.
quest	Input data using AI experience tools Let's try manipulating it.	Simulations explain how changes in AI input values affect output. Can be.	Adjust the input settings in the AI system and get the results can be predicted and compared and analyzed.
understanding	AI classifies by certain criteria (color, size, etc.) I can explain what I am doing.	You can find common features in the data and explain the classification criteria.	AI vectorizes data through feature extraction Can understand and explain the concept of doing.
inference	Why did the AI decide that way? I can speak my mind.	Conditions that influenced AI's judgment I can explain the rules.	AI's decision-making process under given conditions It can be explained or refuted logically.
apply	Applying AI tools to simple problem situations Let's try to solve it.	Solve real-life problems with AI tools, I can explain the application process.	Appropriately applying AI techniques in complex problem situations It can be designed and applied to produce results.
prediction	What results does AI produce with the given data? I guess it will come out.	How AI changes as data changes You can predict how it will react.	New situations based on the model's learning results The results can be predicted and evaluated.
Reflection	When AI makes a mistake, think about why.	You can evaluate the validity of AI results and suggest ways to improve them.	We can comprehensively reflect on AI judgment, human intervention, and ethical issues.

5 AI Understanding Education: What is AI and How Does it Work?

definition

AI understanding education is based on a scientific and structural understanding of "What is AI and how does it work?" and Digitacy. This is an education program that fosters AI-based knowledge and critical thinking skills.

target

- Understanding the concepts, principles, and operating mechanisms of artificial intelligence
- Develop the ability to understand and explain the structure and logic of AI systems.
- Develop a critical attitude that recognizes the limitations and errors of AI technology and reduces blind faith.
- Building foundational capabilities for future collaboration and communication with AI.

Core Competencies

Digitacy-based technological understanding

- Digitacy goes beyond the simple use of digital devices and is the ability to solve real-life problems based on computer science principles and computational thinking.

  It means.

Machine Translated by Google

ÿ AI understanding education focuses on developing technical insight to understand and judge core components such as data, algorithms, and models as information.

ÿ AI thinking and structural analysis capabilities

ÿ Artificial intelligence operates based on the principles of computer science, including science, statistics, mathematics, and logic.

ÿ Learners are trained to structurally analyze causal relationships between causes and effects, repetitive patterns of conditions and effects, and information flow within complex systems.

Through this, we will gain the ability to model and explain how AI works and performs.

ÿ Critical attitude toward AI technology adoption

ÿ The decisions made by AI are not always correct. The AI's decisions depend on various factors such as data bias, algorithm errors, and the designer's intention.

The results may contain limitations and problems.

ÿ Learners should be aware of these possibilities and adopt an attitude of review and questioning rather than uncritically accepting AI results.

ÿ In addition, it is also necessary to have citizenship that can judge for itself and be aware of responsibility for the ethical, economic, and cultural impact of AI technology on humans and society.

are required together.

Applicable Scope and Examples

ÿ Elementary and secondary AI curriculum: Understanding basic AI concepts, algorithm experience activities, AI operation simulations, etc.

ÿ Advanced AI Education for Colleges and High Schools: Understanding Neural Network Principles, Classification/Clustering Algorithm Practice, Model Design Projects, etc.

ÿ AI Literacy Education for the General Public: Analysis of AI Technology Case Studies, Reading and Discussion of AI News, and Raising Awareness of AI Misuse Cases

Training content (example)

Training content (example)	Key Learning Elements
ÿ Understanding the concept of artificial intelligence	- The difference between artificial intelligence and general software - History and development stages of AI Narrow AI vs. General AI - Distinction between weak artificial intelligence (ANI) and strong artificial intelligence (AGI)
ÿ Understanding data and AI	- Overview of how AI works - Fundamental concepts of data science - Linkage between AI and computing systems (hardware, networks, etc.) - Understanding the relationship and role of coding and AI
ÿ Understanding AI algorithms	- Problem solving and exploration algorithms - Knowledge representation and reasoning methods - Data-driven learning: classification, regression, clustering - Basic structure of deep learning

Training content (example)	Key Learning Elements
ÿ AI model development and practice	- Simple data collection and preprocessing - Experience the model design and learning process - Experimental results interpretation and performance evaluation - Experience improving models through feedback

Class activities (examples)

How does artificial intelligence think and judge?
Learning Objectives: Understand the basic structure of how AI receives input and makes decisions.
Watch an animation of the input-processing-output structure of AI (or watch a teacher explain it)
Role-playing game "Creating an AI Doctor" (Symptoms ÿ Diagnosis ÿ Prescription)
Summary: A Simple Comparison of Actual AI Model Structures (Rule-Based vs. Machine Learning-Based)
Use sample classifier simulation sites: Entry, Teachable Machine, etc.

Where will artificial intelligence work?
Learning Objectives Understand that AI operates on computing systems and requires computer resources.
Introduction: "Where will AI work?" (Real-world AI examples, such as YouTube recommendations and smartphone facial recognition)
deployment - Find the AI's body (deduce where the AI operates, the parts contained in each device (computation, storage) - The structure in which computers run AI - Configuring the minimum system to run AI
Summary: "Is AI magic falling from the sky or computation inside a machine?"

How does artificial intelligence learn?
Learning Objectives: Understand the differences between supervised learning, unsupervised learning, and reinforcement learning, and how to apply them.
Case Study: Dog/Cat Classification Problem (Supervised Learning) / Grouping Objects of Similar Colors (Unsupervised Learning)
Learning Structure of AI in Exploration Games ÿ Explanation of Reinforcement Learning Principles
Card Game Learning Style Classification Quiz: Identifying Which Activities Are Learning Types

Let's create your own AI
Learning Objectives: Design and train a simple classification AI.
development ÿ Collect input data (hand gestures, facial expressions, etc.) ÿ Train (label assignment) ÿ Test the model (camera recognition response)
Analyzing the reasons for the discrepancies in the discussion results ÿ Insufficient data? Ambiguous classification criteria?

Machine Translated by Google

Are AI's judgments always correct?
Learning Objectives: Understand AI's judgment accuracy, limitations, and feedback structure.
Analysis of error cases in AI translators (e.g., strange results from translators)
Overfitting vs. Underfitting Explained – "The Over-Memorized vs. the Under-Memorized"
Game AI prediction accuracy improvement game (continuous provision of input-correct answer combinations ÿ performance improvement observed)
organize Conclusion: AI also needs continuous learning

6 AI Collaborative Education: What, How, and Together Can We Do with AI?

definition

AI collaborative education is an education that explores "what", "how", and "together" we can do with AI, and does not utilize AI as a simple tool.

It is a practical education that focuses on the experience of humans and AI solving problems together as cooperative partners beyond the level of human intelligence.

The goal is to foster collaborative thinking that promotes coexistence and symbiosis by fusing creativity, emotions, and intuition with AI's data-based computational capabilities.

target

ÿ Experience human-led problem recognition and collaborative problem-solving with AI.

ÿ Learning that combines human creativity, emotion, and intuition with AI's computational power, speed, and accuracy.

ÿ Develop the ability to utilize and distinguish AI appropriately by experiencing it as a tool and partner.

ÿ Expanding our thinking from "I who utilize AI" to "We who solve problems together with AI."

Core Competencies

ÿ Problem-solving-focused collaboration skills

ÿ Rather than solving problems alone, let humans and AI experience collaborative solutions by leveraging each other's strengths.

ÿ Approach complex problems by integrating AI's analytical and computational power with human emotion and creativity.

ÿ AI-based creative design capabilities

ÿ Develop the ability to conceptualize and design new ideas using AI capabilities.

ÿ A proactive attitude toward human-machine collaboration

ÿ Instead of blindly trusting or rejecting AI, cultivate an attitude of properly understanding and utilizing the potential for collaboration with AI.

ÿ Emphasizes the subjectivity of humans in selecting and coordinating technology.

Applicable Scope and Examples

ÿ Elementary and Middle School Project-Based AI Collaborative Classes: Solving Traffic Problems with AI, Climate Data Analysis Project, and Building Emotion-Recognizing Robots

ÿ CCI (Creative Collaborative Intelligence) Education Practices: Human-Machine Collaborative Play, Communication Experience with Emotion-Recognizing AI, Design Thinking-Linked Classes

ÿ Collaborative AI Practical Training for Universities and Citizens: AI-Based Collaborative Creative Workshop, Collaborative Problem-Solving Hackathon, and Case Studies of Using AI Assistants

Training content (example)

Training content (example)	Key Learning Elements
ÿ Understanding the concept of human-AI collaboration	Human strengths (creativity, intuition, value judgment) vs. AI strengths (computational power, speed, data analysis) - Understanding the concept of collaboration with AI
ÿ Collaborative problem-solving process and design	- A structure where humans recognize problems and AI assists - Problem definition ÿ Role division ÿ Design of collaborative solution process
ÿ Collaborative project-based learning	- Practice with generative AI, recommendation systems, conversational AI, etc. - Projects utilizing AI from a human-centered design (HCD) perspective
ÿ AI as a tool vs. AI as a partner	- Experience the difference between using it as a simple tool and as a collaborative partner. - Clarifying the position of humans as subjects within a cooperative structure.
ÿ Training in human-led AI-powered decision-making	- Experience selecting, rejecting, and reinterpreting AI's suggestions - Includes evaluation of collaborative outcomes and reflective thinking (metacognitive activities).

Class activities (examples)

Where does the problem lie? Can AI help me?
Learning Objectives: Recognize the problems around us and explore the potential of AI to help.
Introduction: Discussion on "What I Want to Ask AI to Do"
Explore Find issues in your school/area (e.g., school trash, safe commute home, classroom noise)
planning What data do you need to solve the problem? Dividing human and AI work
A key shift: Imagine AI as a problem-solving companion, not a tool.

Machine Translated by Google

Let's design a solution with AI.	
Learning Objectives: articulate the role of AI and design a collaborative structure.	
Distinguishing between what analytical AI can do and what humans excel at (table summary)	
Schematic diagram of the division of roles ÿ AI (data collection/classification), I (planning/interpretation/judgment)	
example	Watch the video "Creating a Fine Dust Map with AI" ÿ AI for sensors and analysis, humans for interpretation and campaigns
Envisioning AI as a problem-solving companion, not a tool.	

Let's move together! Solve problems together with AI.	
Learning Objectives: Collaborate with AI tools to implement solutions.	
Project	Classroom Noise Analysis with AI - AI: Receive sound data, measure decibels, and visualize - People: Suggest rules and practice behaviors for a quiet environment
The function of AI as an auxiliary material is to be an execution tool, but it is recognized that humans set the direction.	

"What if AI is better than me?" (Competition)	
Learning Objectives: Compare and emotionally understand the differences and similarities between AI and humans.	
activity	Role-play discussion of watching videos of AI drawing/writing: "Can AI replace me?" Comparing Human and AI Capabilities: The Limits and Potential

"AI, help me!" (Usage)	
Learning Objectives: Recognize and utilize AI as an effective tool.	
activity	Experience real-world AI tools like text summarization, translation, and image classification. [Summary] Creating a "Work That Should Be Left to AI vs. Work That Should Be Done by People" Classification Card A brief introduction to ethical issues

"Let's solve problems with AI" (collaboration)	
Learning Objectives: Gain experience collaborating with AI to solve shared problems.	
activity	Team composition: Design a problem-solving project using AI tools (e.g., creating noise maps, monitoring the environment, analyzing safe routes, etc.) Division of roles: AI and human roles are divided to carry out the project. Sharing Results: "How did our team collaborate with AI?"

"A true team is one that leverages each other's strengths" (Symbiosis)	
Learning Objectives: Understand what AI and humans each excel at and consider coexistence strategies.	
activity	Case Study: Discussing Ethical Issues Arising from AI Interviews, AI Judges, and AI Recruitment Building a Complementary Model: Co-Designing "What Would Human-Centered AI Look Like?" Create your own declaration of human-AI coexistence

"Let's imagine a future where we coexist with AI" (Coexistence)	
Learning Objectives: Design and imagine a future society where we will live with AI.	
activity	Future Scenario Workshop: "What Role Will AI Play in 2050?" After imagining Utopia vs. Dystopia, I presented my desired model of coexistence. My Pledge: "A Human Attitude for Coexisting with AI"

Machine Translated by Google

References

5 Types of Problem Solving by Humans and AI

The changing relationship between humans and AI in problem-solving can be explained in five types.

Stage 1: Competition (AI might replace me)

This is the stage where AI and humans are perceived as competing. At this point, students compare humans to AI, feeling anxious that AI could surpass or replace them. The key concepts are "human vs. AI," comparison, and anxiety. The educational objective at this stage is to critically reflect on the impact of AI technology on humans and understand competing perspectives.

Step 2: Utilization (AI is a tool that helps me)

AI is beginning to be perceived not as a threat, but as a tool to assist in my work. It's a means to increase efficiency and enable automation.

As such, the functions of AI are highlighted. Key concepts include instrumental use, efficiency, and automation. In class, the instrumental value and usability of AI are discussed.

Raise awareness of

Step 3: Collaboration (You can solve problems with AI)

AI is understood as a collaborator (partner) beyond a tool, and is recognized as a being that solves problems together. Humans and AI each share roles and solve problems.

The course focuses on collaborative planning and collaboration to solve problems, with core concepts including role division, collaborative planning, and problem solving. The goal of the course is to foster practical, collaborative thinking.

Step 4: Symbiosis (We must leverage each other's strengths with AI)

This is a step toward recognizing that humans and AI can complement each other by leveraging their respective strengths. Humans possess creativity, ethical judgment, and contextual understanding. AI excels at processing massive amounts of data and rapid analysis. The core concepts are complementing strengths, augmenting roles, and complementary collaboration.

The purpose of this stage of the course is to develop a cooperative strategy between humans and AI to understand each other's strengths and maximize them.

Step 5: Coexistence (Designing the future with AI)

Ultimately, AI is recognized as a partner that can design a sustainable future alongside humans. Along with AI autonomy, human responsibility, and the concept of co-evolution are emphasized. The core concepts are AI autonomy, human responsibility, and co-evolution. The core goal of the course is to envision a sustainable future created by humans and AI together.

<Table> Five Types of Problem Solving by Humans and AI

step	Relationship awareness with AI	Key Concepts	Lesson Objectives
Competitive AI might replace me.		Human vs. AI, comparison, anxiety	Boundaries and Understanding
Utilizing AI is a tool that helps me.		Instrumental use, efficiency, and automation	Recognition of instrumental value
Solve problems with collaborative AI, enabling role sharing, joint planning, and problem solving.			Practical collaboration
AI and I must leverage each other's strengths. Coexistence strategies, ethical design, human-centered technology, and capacity building—recognizing mutually complementary understanding.			
We can design the future with coexisting AI, ensuring AI autonomy, human responsibility, and coevolution.			Envisioning a Sustainable Future

References

Collaboration with AI sparked by generative AI

The advancement of generative AI (Generative AI) goes beyond mere technological advancement; it redefines the collaborative relationship between humans and AI. In particular, GenAI 2.0 is moving away from the tool-centric AI approach of GenAI 1.0 and ushering in a paradigm shift focused on collaboration, where humans and AI collaborate as co-creators and designers, thinking and solving problems together.

GenAI 1.0 was primarily used to generate text, images, and code, with users serving as simple command providers, inputting prompts. While AI at this time provided immediate, static responses, iterative learning was difficult, and continuous, context-sensitive interaction was limited. Because its usage relied on one-time requests (Prompt ÿ Output), it lacked the ability to provide feedback or improve the results.

This was lacking.

In contrast, GenAI 2.0 emphasizes AI as a collaborative partner. Users, too, are transformed from mere inputters to designers and collaborators who define and solve problems alongside AI. AI evolves beyond simply processing user requests to understanding context, accepting feedback, and expanding thinking through meaningful dialogue. This has led to a shift in usage patterns to a continuous interaction model, such as "Prompt ÿ Feedback ÿ Iteration," and responses have also evolved to understand context and discern intent.

These changes are also impacting the expansion of AI's application areas. Beyond simple document creation and translation, GenAI 2.0 can now be applied to high-level knowledge activities and creative tasks such as strategy formulation, educational planning, collaborative creation, and policy development. The initial generative AI, once technology-centric, is now user-centric.

The quality of human interaction, such as UX (user experience), convenience of tool use, and increased productivity, has become more important.

Representative services include GPT-4o, Claude 3, Gemini Advanced, Copilot+, multimodal AI service, MCP, and Agent model.

All of these are moving beyond simple generative functions to become collaborative AI models that continuously evolve based on user feedback.

GenAI 2.0 opens the door to an era where humans go beyond simply 'utilizing' AI and define and solve problems together with AI and create new values.

We've opened up. Now, we're entering a new educational and mindset shift that embraces AI as an intellectual partner, not just a tool.

<Table> Comparison of GenAI 1.0 and GenAI 2.0

division	GenAI 1.0	GenAI 2.0
AI as a Core Concept Content Creation Tool		AI as a Collaborative Partner
User Role Prompt Inputter (Command Provider)		Designers and Collaborators (Co-creation with AI)
AI's role is simple: generator (generating text, images, code, etc.); knowledge partner/creative partner (understanding context, providing feedback, and interactive collaboration);		
Usage method: One-time question-and-answer based (Prompt ÿ Output) Conversational continuous interaction (Prompt ÿ Feedback ÿ Iteration)		
Response characteristics are static, instantaneous, and without iterative learning.		Context-based, continuity-based, intent-based
Automation of application area document creation, summarization, translation, etc.		High-level tasks such as strategy formulation, education, research, and collaborative creation
Model-centric technology-focused (emphasis on model performance)		User-centric (emphasis on UX, tooling, and productivity)
Example service ChatGPT initial version, DALL·E 1		Multimodal, MCP, Agent, etc. GPT-4o, Claude 3, Copilot+, Gemini Advanced, etc.

Machine Translated by Google

7 AI Value Education: How can humans and AI advance toward a valuable society?

definition

AI value education takes as its central question, "How can humans and AI move towards a valuable society?" and examines the impact of AI technology on humans and society.

It is an education that reflects on meaning and seeks ways to preserve human dignity and community values even amidst technological advancements.

target

ÿ Recognize AI as an object of ethical and philosophical reflection, not simply a useful tool.

ÿ Examining the relationship between AI and humans and forming thoughts on the balance between technology and value.

Fostering an attitude that prioritizes human-centered values (dignity, fairness, publicness, etc.) as the standard for AI utilization .

ÿ Presenting a vision to guide AI technology toward pursuing social good.

Core Competencies

ÿ Technical ethics and philosophical thinking

ÿ Develop the ability to think deeply about the social impact and ethical issues of technology.

ÿ Digital citizenship and sense of responsibility

ÿ As a citizen using AI, you must have responsibility and empathy for the community.

ÿ Ability to utilize public interest-oriented technology

ÿ Develop the ability to critically judge and utilize technology to enhance social value.

Applicable Scope and Examples

ÿ Ethics-based AI classes for elementary, middle, and high schools: AI and fairness, personal information and AI, algorithmic bias and responsibility, etc.

ÿ Public and Civic Education: AI for Good, Good AI, Digital Citizenship Education, etc.

ÿ College liberal arts courses: Philosophy of technology, AI ethics, coexistence of humans and machines, etc.

Training content (example)

Training content (example)	Key Learning Elements
ÿ Reflection on the relationship between technology and humans	- Raising fundamental questions about the purpose of technological advancement - Exploring the impact of AI on human life - Introducing the concept of a human-centered technology philosophy
ÿ Understanding the basics of AI ethics and social responsibility	- Concepts of algorithmic ethics (bias, accountability, transparency, etc.) - Data use ethics and privacy protection - Considering human responsibility for AI decisions
ÿ Values centered on human dignity and the public good	- Understanding of values such as fairness, inclusiveness, and publicness - Recognize the need to protect human dignity and rights when utilizing AI.
ÿ Training in technology selection and value judgment	- Analysis of value conflict cases in technological choice situations - Simulation and discussion of ethical dilemmas - Presentation of alternatives and performance of critical thinking activities
ÿ AI for Good practical experience	- Exploring good AI practices (e.g., responding to the climate crisis, assisting people with disabilities) - Planning AI utilization for the socially disadvantaged and minorities Designing a 'Good Life with AI'

Class activities (examples)

Why does AI have 'value'?	
Learning Objectives - What are values? - Why do they only have meaning in human relationships?	
Card discussion on the activity "Who Determines Value?" (AI's 'Usefulness' vs. 'Human Needs')	
Define "AI's value arises from its relationships with humans."	

What is the instrumental value of AI?	
activity	Analysis of various cases that benefit from AI (industrial automation, recommendation algorithms, logistics efficiency, etc.) Understanding the Structure of AI Technology as a "Useful Tool"
A Critical Perspective: Identifying the Limitations and Problems of "Technology for Profit"	

Can AI assist human goals?	
activity	Analysis of a life-or-death scenario (moral dilemma) in autonomous vehicles. "If AI can make its own decisions, who will be held accountable?"
Emphasize that the use of organizing techniques must be accompanied by moral standards.	

What are the values of AI that we should pursue?	
activity	Announcing the creation of an "AI Ethics Charter" or "AI Values Statement": A personal declaration on "What should AI be to humans?"
Discussions on the future of society regarding the creation of expanded self-consciousness and the possibility of autonomous judgment.	

Machine Translated by Google

References

AI values education, not AI ethics education

AI values education must go beyond simply enforcing ethical norms or preventing technological misuse. While education that prevents AI misuse is important, it alone is not sufficient. What we should strive for is "values education," where we collectively reflect on why and how AI should be used, and consider the direction humans and society should pursue together.

"Value" is not an inherent property of a thing or technology itself, but rather the significance that arises from its relationship with humans. Therefore, AI technology itself is merely a neutral tool, neither good nor evil. What matters is how humans handle AI and what kind of relationships they form.
The question is whether to conclude it or not. This is because the same technology can produce completely different results and meanings depending on the context and purpose of its use.

AI value education is an education that helps people view technology as an object with human meaning, and to this end, it is necessary to understand the 'instrumental value' of technology.
It is necessary to distinguish between "purposeful value" and "instrumental value." "Instrumental value" refers to the perception of AI as a helpful tool, a device that solves problems more quickly and conveniently.
This is the perspective. On the other hand, purposive value refers to the attitude of designing and using AI so that it operates for the good of human society. Fair algorithm design, utilization for the socially disadvantaged, and data utilization for sustainability are all examples of practicing purposive value.

From this perspective, AI values education should not only cultivate the ability to use technology instrumentally, but also expand students' thinking, enabling them to imagine and realize a better society through technology. This means that education should go beyond warning about the risks of technology and expand to foster critical reflection and practice to uphold human dignity and community values (such as fairness, publicness, and inclusiveness).

AI values education is not simply about managing the ethical risks of technology. It is a visionary education for a valuable society created by humans and AI together. Through this, we must cultivate not just "people who are good at using AI," but "people who can live well with AI." To achieve this, AI values education must begin with the instrumental value of technology, but ultimately reach its ultimate purpose.

<Table> Instrumental and Purposeful Values

division	explanation
instrumental value	Emphasizes the instrumental role of technology in providing efficiency, speed, convenience, and productivity. Examples: AI-powered production cost reduction, personalized recommendation systems for consumers
Purposeful value	The view that the use of technology should be directed toward intrinsic goals such as human dignity, freedom, safety, justice, and happiness. Example: AI is not meant to replace humans, but rather to help us understand and respect human life more deeply.

References

Machine Translated by Google

References

Bahar Mahmud et al. (2023). A Study of Human–AI Symbiosis for Creative Work: Recent Developments and Future Directions in Deep Learning. ACM Transactions on Multimedia Computing, Communications and Applications, Volume 20(2), 1-21.

Daniels, U et al. (2023). AICOMP-Future Skills in a World Increasingly Shaped By AI. Ubiquity Proceedings, 3(1): 230-239. https://doi.org/10.5334/uproc.91

Daniel, Z. (2013). From Computational Thinking to AI Thinking. IEEE Intelligent Systems, 28(6), 2-4. DOI: 10.1109/mis.2013.141

EU (2022). DigComp 2.2: The Digital Competence Framework for Citizens - With new examples of knowledge, skills and attitudes.

European Commission (2017). DigComp 2.1: The Digital Competence Framework for Citizens with eight proficiency levels and examples of use. Luxembourg: Publications Office of the European Union.

JISC (2014). Developing digital literacy. Retrieved from https://www.jisc.ac.uk/full-guide/developing-digital-literacies

G. G. Lee et al. (2023). Multimodality of AI for Education: Towards Artificial General Intelligence. arXiv:2312.06037v2 [cs.AI].

Gilster, P. (1997). Digital literacy. New York: Wiley Computer Pub.

Latif et al. (2023). AGI: Artificial General Intelligence for Education. https://doi.org/10.48550/ arXiv.2304.12479.

Nataliya Kosmyna et al. (2025). Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task. https://doi.org/10.48550/arXiv.2506.08872

NCEE (2024). Framework for AI-Powered Learning Environments. Washington, DC: National Center on Education and the Economy.

OECD (2009). PIAAC Problem Solving in Technology-Rich Environments (PSTRE).

Paul, R., et al. (2018). AI Thinking for Cloud Education Platform with Personalized Learning. Proceedings of the 51st Hawaii International Conference on System Sciences, 3-12. DOI: 10.24251/HICSS.2018.003

Popstar et al. (2019). elating Artificial Intelligence to the Computational Thinking framework. The Third International Conference on Computational Thinking Education.

Popstar et al. (2019). elating Artificial Intelligence to the Computational Thinking framework. The Third International Conference on Computational Thinking Education.

Ruixun Dai et al. (2024). The roles of AI and educational leaders in AIÿassisted administrative decisionÿmaking: a proposed framework for symbiotic collaboration. The Australian Educational Researcher 52(3).

UNESCO (2018). A Global Framework of Reference on Digital Literacy Skills for Indicator 4.4.2. UNESCO Institute or Statistics, Information Paper No. 51.

UNESCO (2019). Recommendations on Assessment Tools for Monitoring Digital Literacy within UNESCO's Digital Literacy Gollobal Framework. Information Paper No. 56.

UNESCO (2021). Recommendation on the Ethics of Artificial Intelligence. https://unesdoc.unesco.org/ ark:/ 48223/ pf0000381137

UNESCO (2024). AI competency framework for students.

Wanxin Yan et al. (2024). Benefits and Challenges of Collaboration between Students and Conversational Generative Artificial Intelligence in Programming Learning: An Empirical Case Study. Education. Sci. 14(4), 433; https://doi.org/10.3390/educsci14040433.

Yijia Shao et al. (2025). Future ofWork with AI Agents:Auditing Automation and Augmentation Potential across the USWorkforce. https://doi.org/10.48550/arXiv.2506.06576

Ministry of Education (2022). Practical Arts (Technology, Home Economics)/Information Science Curriculum.

Kim, Su-eun (2025). A Case Study on Strengthening AI Literacy in Academic Presentation Classes: Focusing on Self-Correction and Management Skills. Korean. International Literature Forum, 68, 175-207.

Kim Tae-ryeong, Ryu Mi-young, and Han Seon-gwan (2020). A Basic Study on a Framework for Artificial Intelligence Education in Primary and Secondary Schools. Journal of the Korean Society for Artificial Intelligence Education, 1(4). 31-42.

Ryu, Mi-young, Cheon, Jong-pil, and Han, Seon-kwan (2025). A Study on the Ethical Factors of ChatGPT for AI Ethics Education. Journal of the Korean Society for Artificial Intelligence Education. 6(1), 37-46.

Ryu, Mi-young, and Han, Seon-kwan (2022). Development and Application of a Digital-Based Creative Collaborative Intelligence Teaching and Learning Model. Korean Journal of Educational Studies, 42(2), 117-130.

Ryu, Mi-young, and Han, Seon-kwan (2023). The Study on Test Standard for Measuring AI Literacy. Journal of the Korea Computer Information Society, 28(7). 39-46.

Choi, Suk-young (2024). Core competency framework and educational plan for future talent in the era of generative AI. Journal of the Korean Society for Computer Education, 27(9), 23-33.

Choi, Hyeon-jong (2021). A Study on AI Thinking Education Based on Computational Thinking. Journal of the Korean Society for Computer Education, 24(3), 57-65.

Korea Institute of Curriculum and Evaluation (2018) "Analysis of the Current Status of Digital Literacy Education in Curriculum Education and Study on Improvement Measures."

Korea Education and Research Information Service (KERIS) (2023) "2023 Study on Improving Performance-Based Digital Literacy Assessment Tools and Measuring the Level of Elementary and Middle School Students."

Korea Education and Research Information Service (2023) Report on "Digital Literacy Structure and Achievement Standards by Subject"

Korea Education and Research Information Service (2023). Development of a tool to measure AI and digital competency for prospective and current teachers.

Korean Society for Information Education (2023). AI Digital Textbook Elementary School Information Science Curriculum (Based on 34 Lessons by Grade Group).

Han Seon-gwan, Ryu Mi-young, and Kim Tae-ryeong (2021). Artificial Intelligence Education for AI Thinking. Seongandang.

Machine Translated by Google

Korean AI Education Society 2025 Policy Research Report

AI Basic Society, a Great Transformation

in Education: Designing the Future of AI Education 2.0

Issue date: August 2025

Publisher: Korean Society for Artificial Intelligence Education

22851 155-11 Robot Land-ro, Seo-gu, Incheon, Room 1002

Tel: 032) 540-1318 | Fax: 032) 540-0288

www.aied.or.kr

Publisher: Adval, Inc. Tel. 0507-1351-0965

한국인공지능교육학회

인공지능교육연구소